Data Asset Mismatch in the Digital Economy: A Dynamic Investment Perspective

Abstract: In the digital economy, data is a strategic asset, yet its returns often mismatch with infrastructure investment. This paper develops a dynamic framework modeling data as informational capital that improves productivity beliefs through learning. Firms jointly invest in physical and data capital under adjustment costs. Data enhances production by refining knowledge and improving capital allocation. Optimal data accumulation balances user cost against marginal informational gains. The model yields two equilibria: a "data poverty" trap from low marginal returns, and a "data hoarding" equilibrium where persistent value drives excessive accumulation and entry barriers. These findings explain data-asset mismatches in digital economies and contribute to the literature on data economics.

Keywords: Data assets; Data capital accumulation; Data asset mismatch

1. Introduction

In the digital economy era, data has become one of the most critical strategic assets for enterprises. Computing power, emerging as a new key productive force following electricity, thermal energy, and mechanical power, has become the core engine driving the development of the digital economy. The deployment of computing power relies heavily on the support of Internet Data Center (IDC) infrastructure. As the core physical carrier for data storage, computation, and processing, the construction and operational standards of IDCs directly determine the efficiency of developing and utilizing enterprise data assets. With the rapid advancement of information technology, the volume of data accumulated by enterprises has grown exponentially. However, not all of this data possesses tangible commercial value. According to Veritas' "Data Iceberg Report," a significant proportion of data stored by global enterprises falls under the category of dark data—data whose value remains unclear. In Chinese enterprises' data storage, dark data accounts for as much as 54.5%, far exceeding the global average. This phenomenon profoundly reveals the data management dilemma facing contemporary enterprises: a severe mismatch between the effective value of data assets and the actual investment required.

Theoretically, as a key tool for enterprises to predict random outcomes and optimize business processes, data accumulation follows the laws of capital evolution and faces convex adjustment costs (Farboodi et al., 2022). In data investment decisions, enterprises must balance current costs against future informational returns. The precision of knowledge generated from data directly determines the marginal output of physical capital, thereby influencing enterprise scale. Signal precision reflects the quality of an enterprise's data infrastructure, while signal quantity depends on the stock of data capital processed through data mining technologies. Data acquisition and processing infrastructure constitutes a critical component of corporate data capital accumulation, encompassing core elements such as server performance, network bandwidth, and storage capacity. Steady-state analysis indicates that optimal data stock is determined by the equilibrium between user costs and marginal information benefits, giving rise to two heterogeneous phenomena: First, when data mining techniques yield excessively low marginal output in early stages, firms may fall into the “data poverty” trap; Second, when data depreciation rates are low or knowledge persistence is strong, the multiplier effect of information value may lead to “data hoarding,” creating entry barriers.

Regarding the relationship between IDC deployment and corporate data assets, a close correlation exists between IDC deployment and data asset mismatch: when a company's IDC infrastructure is weak and data processing capabilities are insufficient, the marginal output of data mining technology remains low, potentially trapping the company in a “Data Poverty” trap with severe mismatches between computational resource investment and output; Conversely, as data depreciation rates decline and knowledge persistence increases, leading enterprises tend to deploy large-scale IDCs, creating a data hoarding effect. Large IDC clusters reduce unit data processing costs through economies of scale, thereby establishing high entry barriers.

Existing research often treats uncertainty as an exogenous factor, overlooking enterprises' proactive investment in data to enhance productivity forecasting capabilities. Accordingly, this paper contributes to the literature on the economics of data and digital infrastructure in several respects. First, this study develops a theoretical framework that models data as a form of informational capital within firms’ dynamic decision-making process. By incorporating Bayesian learning into the accumulation of data capital, the model highlights how data investment improves the precision of productivity information and indirectly affects firms’ production and capital allocation decisions.

Second, the paper provides a theoretical explanation for the emergence of data asset mismatch by identifying heterogeneous equilibrium outcomes in the process of data accumulation. Depending on the technological characteristics of data mining and the depreciation of data capital, firms may converge to either a “data poverty” equilibrium characterized by insufficient data investment or a “data hoarding” equilibrium characterized by excessive data accumulation.

Third, this study complements the theoretical analysis with empirical evidence based on firm-level data. Using a difference-in-differences framework that exploits the quasi-natural experiment of IDC license acquisition, the empirical results show that the introduction of Internet Data Centers significantly mitigates data asset mismatch within firms. Mechanism analyses further reveal two important channels. On the one hand, IDC infrastructure reduces the cost of data storage and processing, thereby improving the utilization efficiency of data elements and promoting the accumulation of both self-use and transactional data assets. On the other hand, IDC infrastructure enhances firms’ digital innovation capacity, significantly increasing digital patent output as well as the depth and breadth of digital technology application. These findings highlight the critical role of computing infrastructure in facilitating efficient allocation and productive utilization of data assets.

2. Literature Review

The transition of data from a byproduct of operations to a core intangible asset has prompted extensive research into its valuation and economic implications (Veldkamp, 2023). A defining characteristic of data is its nonrivalry, which theoretically allows for infinite replication and increasing returns to scale, yet practically creates strong incentives for firms to hoard data to build competitive moats and preserve privacy (Jones & Tonetti, 2020). This theoretical insight into data hoarding serves as a critical micro-foundation for broader macroeconomic shifts. Specifically, the accumulation of nonrival data by early adopters restricts broader economic efficiency and acts as a barrier to entry, directly contributing to the rise of "superstar firms" that disproportionately capture market share and drive down the aggregate labor share (Autor et al., 2020). Consequently, the nonrival nature of data provides a theoretical basis for the phenomenon of "data hoarding," representing a distinct form of over-investment in the digital economy.

Conversely, the micro-level process by which firms accumulate and utilize this data is fundamentally a mechanism of overcoming information friction. Firms utilize data as signals to continuously update their beliefs regarding unobserved firm-specific productivity, a dynamic analogous to the Bayesian learning frameworks in which firms infer efficiency from realized outcomes (Jovanovic, 1982). However, translating raw data into actionable knowledge is hindered by significant frictions. Empirical evidence in capital dynamics demonstrates that firms face substantial convex adjustment costs when altering their factor inputs, preventing costless and instantaneous scaling (Hall, 2004). Furthermore, in the presence of uncertainty, firms exhibit "wait-and-see" dynamics due to non-convex adjustment frictions, further distorting optimal capital allocation (Bloom, 2009). The realization of value from data, akin to other intangible assets, often necessitates complementary organizational investments, creating a "productivity J-curve" where early-stage returns appear misleadingly low or negative (Brynjolfsson et al., 2021). When applied to data capital, these adjustment costs and complementary investment requirements imply that firms with low initial data processing capabilities may find it optimal to under-invest, leading them to persist in a low-information equilibrium analogous to "data poverty."

Crucially, data hoarding and data poverty are not isolated phenomena but rather divergent evolutionary outcomes stemming from the same theoretical root: the interaction between data's nonrivalry and the severe adjustment costs inherent in information acquisition. In a frictionless environment, a firm's optimal data stock should reach a steady-state equilibrium where the marginal user cost of data equals its marginal information benefit, a condition formalized in the burgeoning literature on data as capital (Farboodi et al., 2022; Veldkamp, 2023). However, these frictions cause actual data accumulation to deviate significantly from this optimal theoretical benchmark, resulting in a structural "data asset mismatch."

While prior literature has developed robust methodologies to measure the adjustment costs and allocation efficiency of physical capital (Hall, 2004), the application of these frameworks to quantify the specific mismatch in data assets remains a critical theoretical and empirical gap. Theoretically, standard capital models lack the non-linear Bayesian updating mechanisms necessary to explain the extreme misallocations of data poverty and hoarding. Empirically, addressing this gap requires a precise measurement of the deviation between a firm's actual data asset accumulation and its expected optimal level. Establishing this theoretical benchmark and empirical measurement not only advances the understanding of digital capital misallocation but also provides the necessary foundation to evaluate how external technological interventions, such as the provision of robust computing infrastructure, might alter the adjustment cost structure and mitigate this underlying mismatch.

3. Data Asset Mismatch

Drawing on Maryam Farboodi, et.al (2022) ' research, we build a framework to describe data asset mismatch, existing in enterprise. We assume data is a kind of tool for firms to forecast random outcomes. High-quality data can help firms optimize business processes. The data accumulation follows the capital evolution equation and faces convex adjustment costs. Firms utilize data to optimize capital and data investment, the precision of knowledge generated by data directly determines the marginal output of physical capital, thereby influencing enterprise scale, while data investment is driven by the trade-off between current costs and future information gains and residual asset value. Steady-state analysis indicates that the optimal data stock is determined by the equilibrium between user costs and marginal information benefits, giving rise to two distinct heterogeneous phenomena: when data mining technologies yield excessively low marginal outputs in their early stages, enterprises may fall into a "data poverty" trap; conversely, when data depreciation rates are low or knowledge persistence is strong, the multiplier effect of information value may lead to "data hoarding" and entry barriers.

The key differences between our model and other capital accumulation model are three-fold: 1) Rather than treading uncertainty as exogenous, firms actively invest in data to improve productivity forecasts; 2) Data capital does not directly produce output; instead, it enhances knowledge precision, which guides physical capital allocation; 3) Beliefs and capital co-evolve through a non-linear Bayesian updating process, generating potential poverty traps and hoarding equilibria absent in standard models.

3.1 Firm Environment

Consider a discrete-time. Infinite-horizon economy populated by a representative firm that combines physical capital and stochastic productivity to produce a homogeneous output. The production technology takes a standard Cobb-Douglas form:

$$Y_t = A_t K_t^{\alpha}, 0 < \alpha < 1$$

Where K_t denotes the physical capital stock installed at the beginning of period t, and A_t represents the true, but imperfectly observed, total factor productivity(TFP). The curvature parameter α captures the output elasticity of capital, implying diminishing returns to capital accumulation.

Drawing on previous researches, the fundamental source of uncertainty in this environment stems form the unobservable nature of productivity. While the firm utilizes capital in production, it does not possess perfect knowledge of the current productivity level A_t. Instead, it must form and update beliefs based on available information. To capture the persistent nature of productivity shocks commonly documented in empirical macroeconomics, we assume that the logarithm of productivity follows an AR(1) process:

$$a_{t+1} = \rho a_t + \theta_t, \theta_t \sim \mathcal{N}\left(0, \sigma_\theta^2\right), |\rho| < 1$$

The persistence parameter ρ governs the degree to which productivity shocks propagate over time, while σ_θ^2 measures the volatility of fundamental innovations. The normality assumption, combined with the linear Gaussian structure, facilitates analytical tractability in the subsequent Bayesian learning framework.

The key informational friction in this model arises from the fact that the firm cannot observe a_t directly. Instead, it possesses a stock of data capital, denoted D_t, which generates noisy signals about the latent productivity state. Each period, the firm extracts a signal from its data asset:

$$s_{it} = a_t + \epsilon_{it}, \epsilon_{it} \sim \mathcal{N}(0, \sigma_\epsilon^2)$$

Where the measurement error ϵ_{it} is independently and identically distributed over time and independent of the fundamental shock θ_t. The precision of this signal, σ_ϵ^{-2}, reflects the quality of the firm's data infrastructure, while the quantity of signals depends on the stock of data capital through a data mining technology h(D_t). This technology captures the idea that larger data volumes enable more precise or more numerous insights about the underlying production environment.

The firm's information set at any point in time is characterized by its posterior beliefs about a_t. For analytical convenience, we work knowledge precision, defined as the inverse of the poster variance:

$$\Omega_t \equiv \Sigma_t^{-1}$$

This precision metric capture the firm's degree of informational awareness: higher Ω_t implies more accurate knowledge of productivity, which in turn enables more efficient capital allocation decisions.

3.2 Data Capital Accumulation

We model data as a from of capital stock that exhibits both durability and investment-driven accumulation. Following the canonical capital accumulation framework. The evolution of data capital is governed by :

$$D_{t+1} = (1 - \delta_D)D_t + I_{D,t}$$

Where D_t denotes the stock of data capital at the beginning of period t, $I_{D,t}$ represents gross investment in data acquisition and processing infrastructure Durgin period t, and $\delta_D \in$ （0， 1） is the depreciation rate of data capital. The depreciation parameter captures the notion that data becomes obsolete or loses relevance over time due to changes in the economic environment , shifts in consumer preferences, or technological advancements that render older data less informative.

A critical departure from standard capital theory lies in the nature of data investment. Unlike physical capital investment, which typically exhibits linear accumulation, data investment entails convex adjustment costs. Specifically, the total cost of investing $I_{D,t}$ units in data capital is given by:

$$C_{INV}(I_{D,t}) = \frac{1}{2}\kappa_D I_{D,t}^2, \kappa_D > 0$$

This convex cost structure captures several realistic features of data accumulation. First, it reflects diminishing returns to rapid data expansion: firms cannot instantly build massive data infrastructures without incurring progressively higher marginal costs due to organizational bottlenecks, privacy compliance burdens, or the scarcity of data engineering talent. Second, it implies that firms have incentives to smooth their data investment over time, as lumpy investment strategies would incur disproportionately high costs. This smoothing motive generates interesting dynamics in the interplay between data investment and physical capital allocation, as firms must balance the marginal benefits of immediate information gains against the convex costs of accelerated accumulation.

While data capital represents the physical infrastructure for storing and processing information, the firm's actual knowledge about its productivity evolves through a Bayesian updating process that incorporates both the natural decay of existing information and the acquisition of new signals from the data stock.

Let $\Omega_t \equiv \Sigma_t^{-1}$ denote the precision of the firm's posterior belief about the latent productivity state. The evolution of knowledge precision follows a non-linear recursive structure derived form the Kalman filter:

$$\Omega_{t+1} = \underbrace{\left[\rho^2(\Omega_t)^{-1} + \sigma_\theta^2\right]^{-1}}_{\text{Information depreciation}} + \underbrace{z_i h(D_t)\sigma_\epsilon^{-2}}_{\text{Learning from data}}$$

The first term captures the natural depreciation of knowledge due to the persistent but evolving nature of the underlying productivity state. Even if the firm receives no new signals, its existing knowledge becomes less precise overtime because the state a_t itself evolves stochastically.

The second term represent the augmentation of knowledge through active learning from the data capital stock. The firm possesses a stock data D_t at the beginning of period t. During the period, it deploys this data to generate informative signals about the current productivity state.

The joint dynamics of (D_t, Ω_t) thus constitute a recursive system where data investment today enhances future knowledge precision, which improves future physical capital allocation decisions, which in turn generates returns that fund further data investment. This feedback loop between real capital accumulation and informational improvement lies at the heart of the model's economic mechanism and distinguishes it from standard frameworks where information is either exogenous or costlessly available.

3.3 Optimization

Given the informational frictions and state variable dynamics described above, the firm's decision problem is inherently dynamic and recursive. At each point in time, the firm must make choices that not only affect current profits but also shape its future information set and productive capacity.

We formulate the firm's problem as a dynamic programming problem with two state variables: the knowledge precision Ω_t and the data capital stock D_t. Both variables are predetermined at the beginning of period t. The firm chooses two control variables each period: the physical capital input K_t (which determines current output) and the data investment $I_{D,t}$ (which determines future data capital).

The firm's objective is to maximize the expected present discounted value of current and future net cash flows. In recursive form, the Bellman equation for this problem is :

$$V(\Omega_t, D_t) = \max_{K_t, I_{D,t}} \left\{ \Pi(K_t, \Omega_t) - \frac{1}{2}\kappa_D I_{D,t}^2 + \frac{1}{1+r}\mathbb{E}_t[V(\Omega_{t+1}, D_{t+1})] \right\} \quad (1)$$

Subject to the transition equations for the state variables:

$$D_{t+1} = (1-\delta_D)D_t + I_{D,t}$$

$$\Omega_{t+1} = \left[\rho^2(\Omega_t)^{-1} + \sigma_\theta^2\right]^{-1} + z_i h(D_t)\sigma_\epsilon^{-2}$$

Where $V(\Omega_{t+1}, D_{t+1})$ denotest the value function, $\mathbb{E}_t$ denotes the expectation conditional on the firm's information set at time t.

The current-period payoff consists of two component. The first term, $\Pi(K_t, \Omega_t)$, represents the firm's expected operating profits given its current knowledge precision and choice of physical capital. The second term, $\frac{1}{2}\kappa_D I_{D,t}^2$,captures the convex adjustment cost of data investment, which is paid upfront in period t. The final term is the discounted expected continuation value, which depends on the future states Ω_{t+1} and D_{t+1} that result from today's choices and the stochastic evolution of the environment.

A critical feature of our model is that the firm does not know its true productivity A_t when making production decisions. Instead, it must base its capital choice on its current beliefs, summarized by the knowledge precision Ω_t. The expected profit function $\Pi(K_t, \Omega_t)$ captures the firm's expected operating surplus given these beliefs:

$$\Pi(K_t, \Omega_t) = P \cdot \mu(\Omega_t)K_t^\alpha - rK_t \quad (2)$$

Where P is the output price, and (Ω_t) is the certainty-equivalent productivity factor that summarizes the firm's effective productive capacity given its informational state.

The recursive formulation reveals the key intertemporal trade-offs facing the firm. At each period, the firm must allocate resources between two competing uses: employing physical capital to generate current output, and investing in data capital to improve future information.

The value function $V(\Omega_t, D_t)$ embodies the firm's maximal expected discounted profits given its current informational and data positions. It is increasing in both arguments: higher knowledge precision improves current and future capital allocation decisions, while a larger data capital stock enables more learning and thus higher future precision.

3.4 Optimality Conditions and Derivations[1]

We begin with the firm’s choice of physical capital K_t, which is static conditional on the current information state. Since capital is fully variable each period with no adjustment costs, the firm solves a static expected profit maximization problem given its knowledge precision Ω_t:

The FOC is obtained by differentiating the current-period payoff with respect to K_t:

$$\frac{\partial}{\partial K_t}\left[\Pi(K_t, \Omega_t) - \frac{1}{2}\kappa_D I_{D,t}^2 + \frac{1}{1+r}\mathbb{E}_t[V(\Omega_{t+1}, D_{t+1})]\right] = 0 \tag{3}$$

Solving for the optimal physical capital stock gives:

$$K_t^* = \left(\frac{\alpha P \mu(\Omega_t)}{r}\right)^{\frac{1}{1-\alpha}} \tag{4}$$

This expression reveals a fundamental relationship between information and real investment. The elasticity of capital with respect to information precision is positive: firms with more accurate knowledge of their productivity operate at a larger scale because they face less uncertainty about the marginal returns to investment. As for economic intuition, information precision acts as a productivity shifter in expectations. When the firm knows its productivity more precisely, it can confidently set capital closer to the ex-post optimal level.

The choice of data investment is inherently dynamic, as it affects future data capital and, through learning ,future knowledge precision. Differentiating the Bellman equation with respect to $I_{D,t}$ yields:

$$-\kappa_D I_{D,t} + \frac{1}{1+r}\mathbb{E}_t\left[\frac{\partial V(\Omega_{t+1}, D_{t+1})}{\partial D_{t+1}} \cdot \frac{\partial D_{t+1}}{\partial I_{D,t}}\right] = 0 \tag{5}$$

From the data accumulation equation $D_{t+1} = (1-\delta_D)D_t + I_{D,t}$:

$$\kappa_D I_{D,t} = \frac{1}{1+r}\mathbb{E}_t[V_D(\Omega_{t+1}, D_{t+1})] \tag{6}$$

Where V_D denotes the marginal value of data capital-the shadow price of an additional unit of data.

This first-order condition embodies a simple yet powerful economic principle: the

[1] The detailed process of solving shows in the appendix

marginal cost of investing in data today equals the discounted expected marginal value of data tomorrow. Data investment is a forward-looking decision. The firm incurs costs today to build data infrastructure that will generate informational benefits in the future. The optimal investment level equates the sacrifice at the margin (the additional cost of another unit of investment) with the expected gain (the increment to future firm value from having more data capital).

To characterize the shadow price V_D that appears in the data investment FOC, we apply the envelope theorem to the value function with respect to the state variable D_t. Differentiating the Bellman equation:

$$V_D(\Omega_t, D_t) = \frac{1}{1+r}\mathbb{E}_t\left[\frac{\partial V(\Omega_{t+1}, D_{t+1})}{\partial \Omega_{t+1}} \cdot \frac{\partial \Omega_{t+1}}{\partial D_t} + \frac{\partial V(\Omega_{t+1}, D_{t+1})}{\partial D_{t+1}} \cdot \frac{\partial D_{t+1}}{\partial D_t}\right] \quad (7)$$

The partial derivatives are obtained from the transition equations. From the knowledge precision evolution:

$$\frac{\partial \Omega_{t+1}}{\partial D_t} = z_i h^{'}(D_t)\sigma_\epsilon^{-2}$$

This term captures how an incremental increase in current data capital enhances future knowledge precision by enabling more or better signals to be extracted. The derivative $h^{'}(D_t)$ reflects the marginal productivity of data in generating information—the slope of the data mining technology function.

From the data accumulation equation:

$$\frac{\partial D_{t+1}}{\partial D_t} = 1 - \delta_D$$

This term represents the fraction of data capital that survives to the next period after depreciation.

Substituting these expressions yields the recursive equation for the shadow price of data:

$$V_D(\Omega_t, D_t) = \frac{1}{1+r}\mathbb{E}_t\left[V_\Omega(\Omega_{t+1}, D_{t+1}) \cdot z_i h^{'}(D_t)\sigma_\epsilon^{-2} + V_D(\Omega_{t+1}, D_{t+1}) \cdot (1 - \delta_D)\right] \quad (8)$$

Equation (8) decomposes the current value of data capital into two distinct components. The first term captures the informational value of data: data enables learning, which improves future knowledge precision, and higher precision increases firm value through better capital allocation decisions. The second term captures the asset value of data: data capital that survives to the next period retains its capacity to generate future information, much like physical capital retains its productive capacity. This dual nature—data as both an information input and a durable asset—distinguishes it from standard capital goods.

We similarly apply the envelope theorem with respect to the knowledge precision state Ω_t, Differentiating the Bellman equation:

$$V_\Omega(\Omega_t, D_t) = \frac{\partial \Pi(K_t^*, \Omega_t)}{\partial \Omega_t} + \frac{1}{1+r}\mathbb{E}_t\left[V_\Omega(\Omega_{t+1}, D_{t+1}) \cdot \frac{\partial \Omega_{t+1}}{\partial \Omega_t}\right]$$

The first term captures the direct effect of higher precision on current expected profits. From the expected profit function, this derivative is positive: better information improves current capital allocation efficiency, raising expected output for any given capital stock. The second term involves the propagation of knowledge through time. From the knowledge precision transition, we have:

$$\frac{\partial \Omega_{t+1}}{\partial \Omega_t} = \frac{\rho^2}{\left(\rho^2 + \sigma_\theta^2 \Omega_t\right)^2}$$

This derivative, which we denote A_t, measures the persistence of knowledge precision. Higher current precision leads to higher future precision, even without new signals, because the Bayesian updating process carries forward past information.

Substituting, we obtain:

$$V_\Omega(\Omega_t, D_t) = \frac{\partial \Pi}{\partial \Omega_t} + \frac{1}{1+r}\mathbb{E}_t[V_\Omega(\Omega_{t+1}, D_{t+1}) \cdot \Lambda_t]$$

This recursive equation reveals that knowledge precision is a durable asset in its own right. Current information not only improve today's decisions but also persists into the future, enhancing the quality of all subsequent decisions. This persistence creates a knowledge multiplier effect: the value of acquiring information today is amplified because it continues to pay dividends in future periods through better-informed choices.

Combining the first-order condition for data investment with the envelope conditions yields a unified Euler equation that characterizes the optimal dynamic path of data accumulation. Starting from the investment FOC(6) at time t:

$$\kappa_D I_{D,t} = \frac{1}{1+r}\mathbb{E}_t[V_D(\Omega_{t+1}, D_{t+1})]$$

Now advance this condition one period forward and substitute into the envelope condition for V_D at time t+1. Alternatively, we can substitute the recursive expression for V_D directly into the current FOC to obtain a forward-looking equation that links current investment to future marginal conditions.

After substitution and rearrangement, we arrive at the core Euler equation:

$$\kappa_D I_{D,t} = \frac{1}{1+r}\mathbb{E}_t\left[V_\Omega(\Omega_{t+1}, D_{t+1}) \cdot z_i h^{'}(D_t)\sigma_\epsilon^{-2} + \kappa_D I_{D,t+1}(1-\delta_D)\right]$$

This equation embodies the complete set of intertemporal trade-offs governing data investment. The first term represents the discounted expected informational return on data investment. An additional unit of data capital today generates $z_i h^{'}(D_t)\sigma_\epsilon^{-2}$ units of additional knowledge precision tomorrow, and each unit of precision is valued at the shadow price V_Ω. The second term represents the asset continuation value. Data capital that remains after depreciation retains its capacity to generate future informational returns, and the cost of replacing that capital is saved.

Data accumulation is not merely a technical process but an economic decision governed by the same intertemporal optimization principles that apply to physical capital, albeit with the added complexity that data generates value indirectly through learning rather than directly through production.

3.5 Steady State Analysis

We define a steady state as a situation in which all endogenous variables are constant over time. Formally, a steady-state equilibrium consists of a data capital stock D^*, knowledge precision Ω^*, physical capital stock K^*, data investment flow I_D^*, and shadow prices V_{D^*} and V_{Ω^*} such that:

$$D_{t+1} = D_t = D^*, \Omega_{t+1} = \Omega_t = \Omega^*, I_{D,t} = I_D^*, K_t = K^*$$

And all optimality condition derived in 3.4 are satisfied simultaneously. The steady state abstracts from transitional dynamics and focuses on the long-run equilibrium toward which the firm converges in the absence of further shocks.

We impose time invariance on the transition equations for the two state variables. For **data capital accumulation,** from equation(1), setting $D_{t+1} = D_t = D^*$ yields:

$$D^* = (1 - \delta_D)D^* + I_D^*$$

Solving for steady-state data investment:

$$I_D^* = \delta_D D^*$$

This condition is intuitive: in steady state, data investment exactly offsets depreciation, maintaining the data capital stock at a constant level. The flow of new data investment equals the depreciation flow, analogous to the standard capital accumulation condition in neoclassical growth models.

For **Knowledge precision evolution,** we impose $\Omega_{t+1} = \Omega_t = \Omega^*$ on the Bayesian updating equation gives:

$$\Omega^* = \left[\rho^2(\Omega^*)^{-1} + \sigma_\theta^2\right]^{-1} + z_i h(D^*)\sigma_\epsilon^{-2}$$

This equation is fundamental to understanding the steady-state information environment. It states that the long-run knowledge precision is determined by a balance between two opposing forces: the depreciation of information due to the evolving nature of productivity (the first term) and the acquisition of new information through data-driven learning (the second term).

Rearranging Ω^* yields an expression that isolates the contribution of data capital to steady-state knowledge:

$$z_i h(D^*)\sigma_\epsilon^{-2} = \Omega^* - \left[\rho^2(\Omega^*)^{-1} + \sigma_\theta^2\right]^{-1}$$

The right-hand side represents the "informational gap" that data must fill—the difference between the desired steady-state precision and the precision that would persist from the past in the absence of new signals.

From the FOC condition for physical capital, steady-state capital is:

$$K^* = \left(\frac{\alpha P \mu(\Omega^*)}{r}\right)^{\frac{1}{1-\alpha}}$$

This expression makes explicit the dependence of long-run firm size on the steady-state information environment.

The recursive equations for the shadow prices of knowledge and data simplify considerably in steady state. From the envelope condition for knowledge precision, imposing time invariance yields:

$$V_\Omega^* = \frac{\partial \Pi^*}{\partial \Omega} + \frac{1}{1+r} V_\Omega^* \cdot \Lambda^*$$

The first term denotes the derivative of expected profit with respect to knowledge precision evaluated at steady state, and Λ^* is the steady-state value of the persistence derivative. Solving for V_Ω^*:

$$V_\Omega^* = \frac{\partial \Pi^* / \partial \Omega}{1 - \frac{\Lambda^*}{1+r}}$$

This expression reveals the knowledge multiplier effect in its starkest form. Similarly, from the envelope condition for data capital, steady state implies:

$$V_D^* = \frac{1}{1+r}\left[V_\Omega^* \cdot z_i h'(D^*) \sigma_\epsilon^{-2} + V_D^* \cdot (1 - \delta_D)\right]$$

Solving for V_D^*:

$$V_D^* = \frac{V_\Omega^* \cdot z_i h'(D^*) \sigma_\epsilon^{-2}}{r + \delta_D}$$

This equation expresses the steady-state shadow price of data as the present value of the marginal informational benefits it generates. The numerator, $V_\Omega^* \cdot z_i h'(D^*) \sigma_\epsilon^{-2}$, is the incremental value of information produced by an additional unit of data. The denominator, $r + \delta_D$, is the user cost of data capital. The marginal value of data equals the discounted stream of future informational rents it generates, where the discount rate incorporates both time preference (r) and the depreciation of the asset (δ_D). This parallels the valuation of physical capital, but with the crucial distinction that the "dividend" from data is not direct output but improved information precision.

Combining the steady-state expressions with the FOC condition for data investment yields the core condition that determines optimal long-run data accumulation. Substituting the steady-state shadow price V_D^* into the investment FOC evaluated at steady state:

$$\kappa_D I_D^* = \frac{1}{1+r} V_D^*$$

Using $I_D^* = \delta_D D^*$, we get:

$$\kappa_D \delta_D (r + \delta_D) D^* = \frac{1}{1+r} \cdot \frac{\partial \Pi^* / \partial \Omega}{1 - \frac{\Lambda^*}{1+r}} \cdot z_i h'(D^*) \sigma_\epsilon^{-2} \tag{9}$$

This formidable-looking equation admits a clean economic interpretation. $\kappa_D \delta_D (r + \delta_D) D^*$ represents the marginal user costs of data capital. This equation encapsulates the fundamental trade-off governing long-run data accumulation. The firm expands its data capital until the incremental cost of holding another unit of data exactly equals the incremental benefit that data generates through improved knowledge precision and, consequently, better capital allocation decisions.

3.6 Comparative statics and Economic interpretations

The steady-state conditions yield testable predictions about how structural parameters affect long-run outcomes:

Effect of data depreciation ($\boldsymbol{\delta_D}$): Higher depreciation increases the user cost of data, reducing the optimal data stock D^*. This, in turn, lowers steady-state knowledge precision Ω^* and physical capital K^*. Industries with rapidly obsolescing data—such as those subject to frequent technological or preference shifts—will exhibit lower equilibrium data intensity.

Effect of learning efficiency ($\boldsymbol{z_i}$): An increase of the firm's ability to extract information from data-raises the marginal information benefit of data, shifting the right-hand side of (9) and increasing D^*. This captures the intuitive idea that firms with better analytical capabilities accumulate more data in equilibrium.

Effect of fundamental uncertainty ($\boldsymbol{\sigma_\theta^2}$): Higher volatility in the underlying productivity process increases the rate at which information depreciates, reducing Ω^* for any given D^*. This captures the intuitive idea that firms with better analytical capabilities accumulate more data in equilibrium.,

Effect of persistence ($\boldsymbol{\rho}$): Greater persistence in productivity shocks slows information depreciation, raising Ω^* for any given D^* and increasing the knowledge multiplier Λ^*. Both effects increase the marginal value of data and encourage accumulation. Firms in stable, persistent environments invest more in data because the information they generate remains valuable for longer.

3.7 Extensions: Data Poverty and Data Hoarding

The steady-state analysis reveals how heterogeneous firm outcomes emerge naturally from the model's structure, particularly when the data mining technology $h(D)$ exhibits non-linearities.

Data poverty refers to a situation where the firm's equilibrium data stock is zero or negligible, leaving it with low knowledge precision and small physical capital scale. This outcome arises when the marginal informational benefit of data falls persistently below the user cost for all positive levels of D

The mechanism operates through the S-shaped learning curve $h(D)$, where $h'(D)$ approximately closes to 0, at very low data levels. Firms with minimal data generate negligible informational returns from investment, yielding no meaningful improvement in knowledge precision. Low precision, in turn, depresses physical capital productivity and profits, leaving insufficient resources to fund data accumulation. Such firms remain trapped in a low-information equilibrium, unable to initiate the digital transformation process.

Data hoarding describes the opposite extreme, where firms accumulate massive data stocks, generating persistent competitive advantages that may entrench market power. This outcome emerges when the user cost of data is exceptionally low or when the knowledge multiplier effect amplifies information value dramatically.

Two conditions drive data hoarding. First, low user cost: when the depreciation rate δ_D approaches zero—data retains its value indefinitely—the marginal cost of holding data becomes minimal, supporting very large equilibrium D^*.

Second, high knowledge multiplier: when knowledge persistence Λ^* is close to unity, the denominator $1-\frac{\Lambda^*}{1+r}$ becomes small. Massively amplifying the marginal value of precision V_Ω^*. Each unit of information generated by data yields a present value far exceeding its immediate contribution to profits.

4. IDC

According to Equation (9) in Section 3, the scale of an enterprise's digital assets is determined by the equilibrium between the marginal cost of users and the marginal benefit of information. While enterprise digital transformation mitigate information mismatches and curb efficiency losses by enhancing data scale and quality and improving the granularity of information processing, their fundamental role remains in reducing information asymmetry among entities. Constrained by traditional information systems' lack of inductive, deductive, reasoning, and judgment capabilities (Aghion et al., 2017), corporate decision-making still relies on human value judgments. Data functions merely as an information carrier, struggling to independently become a direct production factor participating in the production process. However, when computing power and algorithms become deeply embedded in data processing, data provides the foundation for value, algorithms determine the value orientation, and computing power empowers enterprises to extract data value. The convergence of these three elements enables computing systems to predict, evolve reasoning, and even replace humans in making comprehensive judgments and complex decisions. This intelligent evolution propels data into becoming a core production factor, deeply participating in the enterprise's value creation process.

As the core platform supporting corporate IT infrastructure, Internet Data Centers (IDCs) provide centralized physical space, redundant power, efficient cooling, and high-speed networking. They build specialized service systems for server hosting, data storage, and operations management—essentially functioning as “computing power hubs” designed to address the high costs, low stability, and operational challenges of self-built data centers. At the level of computing power optimization and data asset allocation, IDC drive the transformation of corporate data assets from “passive storage” to “active value enhancement.” Under traditional IT architectures, data often resides in fragmented storage states with low utilization rates, becoming “dormant assets.” Leveraging centralized computing power supply and intelligent scheduling mechanisms, IDC significantly boost data processing efficiency and retrieval speed, enabling enterprises to unlock data value at lower marginal costs.

An IDC license not only serves as the threshold for enterprises to access critical computing infrastructure but also signifies a leap in their capabilities regarding standardized data management, security compliance, and processing efficiency.

Holding an IDC license indicates that an enterprise possesses a compliant and efficient computing center, significantly enhancing the efficiency of converting raw data into predictive signals. Based on Equation (3), the learning efficiency of enterprises with an IDC license is:

$$z_i(IDC) = z_i^0 \cdot (1 + \theta \cdot IDC), \theta > 0$$

Where, θ denotes the computing power gain delivered by IDC.

The standardized architecture of IDC facilitates data cleansing and structuring, altering the shape of h(D). For small-scale data, enterprises with IDC can more rapidly reach the threshold for increasing marginal returns. Concurrently, IDC are high-energy-consumption facilities with stringent compliance requirements. Obtaining a license entails additional operational energy costs and regulatory compliance expenses for enterprises, reflected in the normalization of the effective depreciation rateδ_D and marginal adjustment cost k_D.

$$\delta_D(IDC) = \delta_D^0 + \gamma \cdot IDC, \gamma > 0$$

According to the FOC in Equation (9), the equilibrium equation for the optimal data asset scale D^* of an enterprise is:

$$\underbrace{\kappa_D \delta_D \frac{r + \delta_D}{1 + r} D^*}_{MC} = \underbrace{\frac{\partial \Pi^* / \partial \Omega}{1 - \frac{\Lambda^*}{1 + r}} \cdot \frac{z_i h^{'}(D^*)}{\sigma_\epsilon^2}}_{MB} \tag{10}$$

When enterprises face data scarcity, their MB curve falls below the MC curve near D approaching zero due to excessively low z_i or prohibitively high technical barriers, leading them to forgo investment. At IDC=1, z_i increases, causing the MB curve to shift upward overall. Enterprises previously unable to cross the investment threshold due to insufficient computing power now see their MB curve intersect with the MC curve in the positive range, empowered by IDC. This shift prompts them to transition from zero investment to optimal-scale investment, thereby alleviating insufficient data investment.

When the depreciation rate δ_D is extremely low or data stickiness is excessively strong, firms will endlessly push data for minimal marginal gains. Energy consumption constraints and compliance requirements imposed by IDC increase δ_D, steepening the MC curve and raising the cost of holding redundant data. As $z_i$improves, firms can achieve the same knowledge precision with fewer high-quality data assets.

Based on this: This paper proposes the following core research hypotheses

H1: Optimizing computing power deployment can significantly alleviate data asset mismatches.

H1a: Computing power deployment promotes the overcoming of technical bottlenecks by enhancing computational efficiency, thereby increasing the accumulation of effective data assets.

H1b: Optimize computing power deployment by strengthening energy efficiency cost constraints and enhancing asset conversion efficiency, thereby correcting disorderly expansion practices.

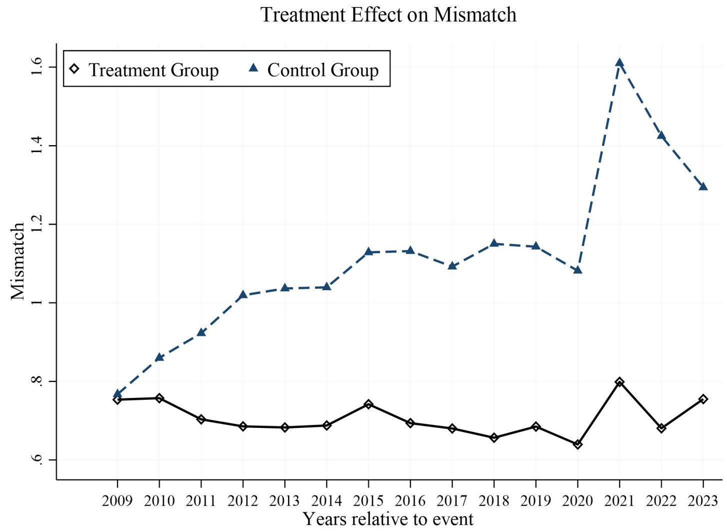


Figure 1 The trend of Mismatch

Figure 1 shows that the degree of data asset mismatch in the treatment group consistently remained lower than that in the control group. The post-policy treatment group maintained stability, while the control group experienced a significant increase in mismatch, with the gap between the two groups widening markedly. This indicates that the pilot policy may have enhanced data asset management capabilities, making the treatment group more resilient to external shocks and thereby relatively improving the mismatch issue.

5. Data and research design

5.1 Sample construction

This paper views a company's possession of an IDC license as an indicator of its optimization of data storage and advancement of computing power infrastructure. So, we manually collected data on IDC operating licenses issued from 2001 to March 2023 from the Ministry of Industry and Information Technology’s Comprehensive Management Information System for Telecommunications Services Market. We set the sample period from 2008 to 2023 since companies disclosed information regarding data assets in the appendix to their annual reports and the format for disclosing annual reports has changed in 2007. We have matched the issuance of IDC licenses with A-share listed companies and processed them as follows: 1) Exclude samples from the financial industry; 2) Exclude samples with negative net worth;3) Exclude samples from the IPO year; 4) Exclude samples with missing key variables. Finally, we obtained data from 4,444 enterprises, comprising a total of 39,102 observations. Corporate data assets are sourced from annual report disclosures, while control variable data is sourced from the CSMAR database.

5.2 Measuring data asset mismatch

Our research methodology follows a methodological framework that draws inspiration from the study conducted by Pedro, et.al (2025). Specifically, we estimate expected data asset using a comprehensive set of firm-specific variables that explain the data investing decisions:

$$\begin{gathered}\text{Exp\_data}_{i,t} = \alpha + \beta_1 \text{ Growth}_{i,t} + \beta_2 \text{ Growth}_{i,t-1} + \beta_3 \text{ ROA}_{i,t} + \beta_4 \Delta \text{ ROA}_{i,t} + \beta_5 \Delta \text{ ROA}_{i,t-1} + \beta_6 \text{ TobinQ}_{i,t} \\ + \beta_7 \text{ Size}_{i,t-1} + \beta_8 \text{ QuickRatio}_{i,t-1} + \beta_9 \Delta \text{ QuickRatio}_{i,t-1} + \beta_{10} \Delta \text{ QuickRatio}_{i,t} + \beta_{11} \text{ Lev}_{i,t-1} + \beta_{12} \text{ Data\_asset}_{i,t-1} \\ + \lambda_j + \varepsilon_{i,t}\end{gathered} \tag{11}$$

Where i and t refer to the firm i and year t. Exp_data is the estimator of firms' data asset based on features of corporate finance, profitability, and data asset investment from the previous period. We include industry fixed effects λ_j to control for unobserved industry characteristics impacting data asset investment in each year.

Then, we calculate Mismatch as the absolute difference between the firm's actual and expected investment as shown in Eq (16):

$$\text{Mismatch}_{i,f} = |\text{ Actual data asset}_{i,f} - \text{ Expected data asset}_{i,f}| \tag{12}$$

5.3 Variable selection and definition[2]

1）Dependent variable: Data asset mismatch. The specific calculation method is shown in Section 5.2. Lower (or higher) levels of Mismatch indicate greater (or less) degree of data asset mismatch.

2）Independent variable: IDC. Drawing on Xu, et.al(2025), we utilizes the "List of Issued Value-Added Telecommunications Business Licenses" published in each batch of fixed announcements from the Comprehensive Management Information System for the Telecommunications Business Market on the Ministry of Industry and Information Technology's government service platform, along with its query system, to obtain relevant information regarding IDC licenses. When applying for IDC services, enterprises are required to complete system evaluations prior to service activation. This means that the prerequisite for service activation is passing evaluations conducted by the approving authority on the operational security systems, website registration systems, resource management systems, and information security management systems of the physical data center. Therefore, holding an IDC operating license can be regarded as the enterprise possessing operational computing resources. We manually collected company names and matched them with precise Unified Social Credit Codes from the Tianyancha database. It determined the issuance year based on license numbering rules, filtered entries containing "Internet Data Center Services" according to business categories, and identified the cities where actual operational data centers are located based on coverage scope. This process yielded a foundational dataset for IDC operating licenses. Based on this, when a listed company itself or any company within its shareholding chain holds an IDC operating license, the variable

[2] The detailed definitions of variables are provided in the appendix

IDC is set to 1; otherwise, it is set to 0.

3）Control variable: Drawing on Giannetti et al.(2015)， we control Company Size (SIZE), Debt-to-Asset Ratio (LEV), Return on Assets (ROA), Company Growth Potential (TobinQ) and other variables.

5.4 Empirical specification

To examine the impact of IDC license on data investment mismatch, we constructed difference-in-difference model:

$$Mismatch_{i,t} = \alpha_0 + \beta_1\ IDC_{i,t-1} + \gamma_1 Controls_{i,t-1} + \sigma_i + \delta_t + \varepsilon_{i,t}$$

Where $Mismatch_{i,t}$ measures the degree of data asset misalignment within an enterprise; $IDC_{i,t-1}$ reflects whether an enterprise possesses an IDC license; $Controls_{i,t-1}$ refers to a set of control variables encompassing characteristics such as a company's finances and board of directors; σ_i denotes the firm fixed effects; δ_t denotes the year fixed effects; $\varepsilon_{i,t}$ denotes error term. Additionally, all company-level regression standard errors in this paper are clustered at the company level.

6. Empirical results

6.1 Summary statistics

Table 1 provide an overview of the summary statistics derived form the sample. The mean value of data asset mismatch (Mismatch) is 0.717, with minimum and maximum values of 0.00163 and 4.499, respectively, indicating significant variations in data asset mismatch across different enterprises. The average IDC license value stands at 0.0273, indicating that enterprises holding IDC licenses remain a minority. Such a distribution suggests that IDC licensing acts as a scarce resource in the digital economy, potentially creating competitive advantages for licensed firms in computing power, data storage, and cloud service provision.

Table 1 Summary statistics

	(1)	(2)	(3)	(4)	(5)
VARIABLES	N	mean	sd	min	max
Mismatch	39,102	0.717	0.529	0.00163	4.499
IDC	39,102	0.0273	0.163	0	1
Size	39,102	22.15	1.330	15.58	28.70
Lev	39,102	0.418	0.209	0.00708	1.262
ROA	39,102	0.0410	0.0755	-1.859	1.285
Cashflow	39,102	0.0467	0.0760	-1.077	0.876

INV	39,102	0.142	0.133	0	0.943
FIXED	39,102	0.208	0.160	0	0.971
Growth	39,102	3.603	634.7	-1.445	134,607
Board	39,102	2.121	0.202	0.693	2.890
Dual	39,102	0.292	0.455	0	1
TobinQ	39,102	2.041	2.233	0.611	259.1
FirmAge	39,102	2.897	0.370	0	4.190
INST	39,102	0.439	0.248	0	1.011

6.2 Realistic characteristics of data mismatch

6.2.1 Time trend

We categorize data asset mismatches into data hoarding and data poverty, plotting their respective time trends based on annual averages. Figure 1 shows the results.

Overall, both exhibit pronounced asymmetric evolutionary characteristics. The degree of data hoarding (red) shows a sustained downward trend throughout the sample period, closely linked to China's digital infrastructure expansion and regulatory refinement: the Broadband China strategy (2013) and subsequent "speed-up and fee-reduction" policies reduced storage costs, curbing indiscriminate expansion, while the Data Security Law heightened compliance awareness, effectively restraining disorderly hoarding.

However, data hoarding rebounded markedly in 2023. This anomaly likely reflects statistical changes following clarification of data asset disclosure policies. In August 2023, the Ministry of Finance issued the Interim Provisions on Accounting Treatment for Enterprise Data Resources, establishing recognition standards for data assets effective January 2024. Consequently, enterprises proactively inventoried previously unreported data resources in 2023, passively increasing book data accumulation. Thus, the observed rise primarily reflects data asset exposure rather than uncontrolled investment — an interpretation consistent with deepening data poverty, as disclosure requirements widened the gap between data-rich and data-poor firms.

In contrast, data poverty (blue) evolved through distinct phases. During 2009-2014, poverty steadily climbed due to lag effects between digital infrastructure deployment and enterprise investment conversion. The 2015-2020 period saw sustained decline, driven by mature cloud technology, reduced costs, and industrial internet applications alleviating cost-constrained poverty. A sharp surge in 2021 resulted from three overlapping factors: diverging digital capabilities under pandemic pressure, heightened compliance thresholds, and rising computing costs from the global chip shortage. The gradual decline during 2022-2023 reflects relief policies like the "East Data, West Computing" initiative and enterprises' progressive adaptation to compliance frameworks.

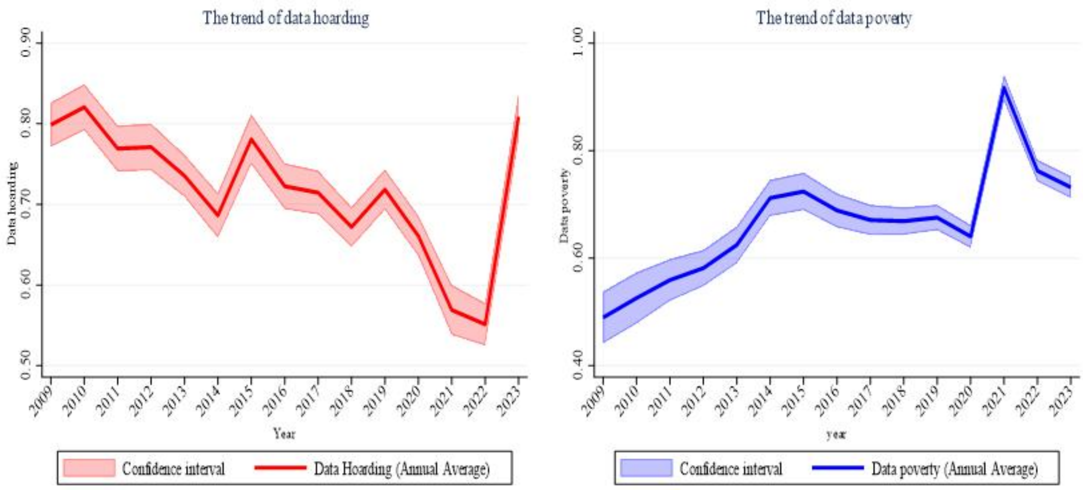


Figure 2 Time trend

6.2.2 Industrial distribution

We have mapped the industry distribution of data asset mismatches according to industry classification (major groups). Figure 2 shows the results. From an industry perspective, the degree of data asset mismatch exhibits significant sector-specific heterogeneity. The information transmission, software, and information technology services sector (I) shows the highest average mismatch, substantially outpacing other industries. This aligns with its status as a core producer of data elements — active data investment and innovation inherently involve higher trial-and-error risks and volatility. Real estate (K) and public administration, social security, and social organizations (O) follow closely, reflecting heightened sensitivity to data compliance and governance, where investment decisions are more susceptible to policy fluctuations. Traditional sectors like mining (B) and manufacturing (C) exhibit moderate mismatch levels, currently undergoing accelerated digital transformation with frequent investment adjustments. Sectors like agriculture, forestry, animal husbandry, and fisheries (A) and construction (E) exhibit lower mismatch levels, potentially due to relatively limited data input and more cautious investment decisions. This distribution reveals that addressing data asset mismatch requires tailored approaches: data-intensive industries need regulatory guidance, while traditional sectors require enhanced capacity building.

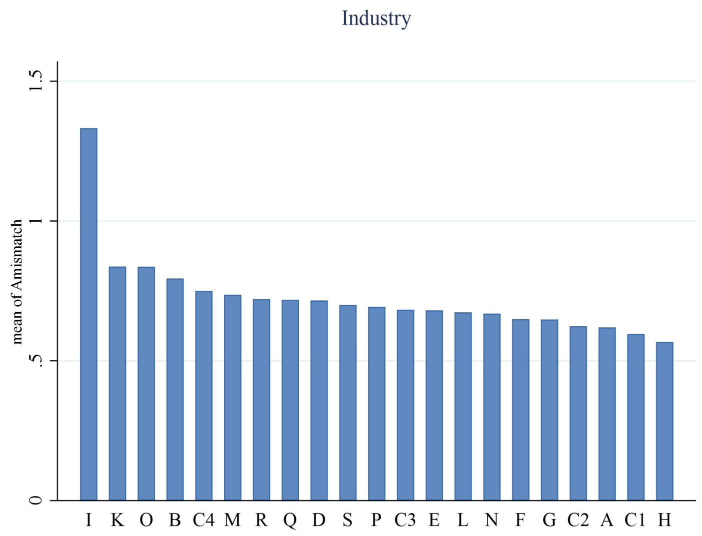


Figure 3 Industrial distribution

6.3 The impact of IDC on data mismatch

First, we test the main hypotheses(H1). We evaluate how enterprise computing power development alleviates the degree of data mismatch. The results of baseline regression are shown in Table (5). Column(1) is the result of adding firm and year fixed effects without control variables. Column(2) adds control variables to the basis of Column(1). The coefficient for IDC is -0.105 and is significantly negative at the 5% significance level. The results show that enterprises holding IDC licenses exhibit an average reduction of approximately 10.5% (or precisely 9.97%[3]) in data asset mismatch compared to unlicensed enterprises. This effect is equally significant in economic terms. Indicating that IDC licensing effectively mitigate distortions in corporate data resource allocation. Consequently, enterprises' data investment levels align more closely with their optimal state determined by their inherent characteristics.

Column(3) and Column(4) represent data hoarding(The actual value of data assets is greater than the predicted value) and data poverty(The actual value of data assets is less than the predicted value) ,respectively. In data hoarding ,the coefficient for IDC is significantly negative. However, it is not significant in data poverty. This asymmetric outcome reveals the limitations of the IDC license. A possible explanation lies in the license's inherent nature as a ‘regulatory’ policy tool. By establishing compliance thresholds for energy consumption, security, and other factors, it raises the marginal cost of blindly expanding data assets, thereby effectively curbing data hoarding driven by agency issues or excessive optimism. However, for enterprises facing data poverty due to financing constraints or technological barriers, the license itself does not provide direct financial support or technological empowerment. Consequently, it struggles to

[3] Calculate as $e^{-0.105}$-1

fundamentally stimulate their investment willingness.

Table 2 Baseline regression

Variable	(1)	(2)	(3)	(4)
	Mismatch	Mismatch	Mismatch	Mismatch
IDC	-0.101*	-0.105**	-0.137***	-0.102
	(0.0517)	(0.0514)	(0.0387)	(0.0737)
_cons	0.00303**	-0.577*	0.490	0.0476
	(0.00152)	(0.319)	(0.309)	(0.308)
Controls	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	38639	38639	18723	18676
adj. R2	0.527	0.528	0.555	0.152

Note:*、**、*** indicate significance at the 10%, 5%, and 1% significance levels, respectively. The same applies below.

6.4 The Parallel Trends Assumption

6.4.1 Pre-Treatment Parallel Trends Test

The parallel trends assumption is a core assumption in DID estimation. Based on the event study approach, this paper examines whether there are pre-existing trend differences between the treatment and control groups, and constructs the following event study model to test for pre-treatment parallel trends:

$$Mismatch_{it} = \beta_0 + \sum_{l=-5}^{-2} \mu_l D_{it}^l + Current_{it}^l + \sum_{l=1}^{5} \beta_l D_{it}^l + \alpha X_{it} + \sigma_i + \delta_t + \varepsilon_{it} \qquad (17)$$

Among them, D_{it}^l denotes the relative time dummy variable, expressed as an indicator function that the value 1 when the condition is met and 0 otherwise4.

μ_l represents the pre-event estimated coefficient, β_l represents the post-event estimated coefficient, and the remaining specifications are the same as in the baseline regression.

According to Zhang & Huang (2023), When the three conditions: the parallel trends assumption, the homogeneous treatment effects path and the no anticipatory effects assumption are satisfied, each point estimate in the event-study plot represents the difference between the current period’s average treatment effects, ATT_l, and the baseline period’s ATT_{-1}, which is excluded from the regression model.

[4] For instance, when individual *i* in period *t* is three periods aways from the treatment(*i.e.*,*l*=-3), the variable D_{it}^{-3} takes the value 1, while all other relative-time dummy variable D_{it}^{l} (where $l \neq -3$) are set to 0. Note that this study defines the base period as the period immediately before the event (pre-event period 1). Therefore, the relative-time variable *l*=-1 is excluded from Equation(2)

As shown in Figure 1, we sequentially conducted pre-trend tests for the data mismatch. The pre-event coefficients are all statistically insignificant, while the post-event coefficients are significant in most periods. Therefore, the pre-trend tests are considered passed. We are also reasonably confident that the post-event trends remain parallel under the counterfactual framework.

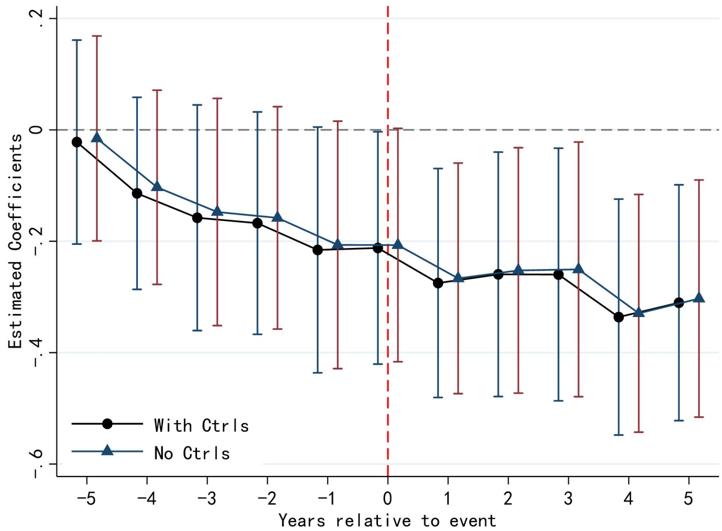


Figure 4 Parallel Trends Test

6.4.2 Robustness Test for Pre-Trends

We employ binning for relative time dummy variables during pre-event parallel trend testing. To further examine whether pre-event trends are parallel, it adopts an interpolation estimation method from Borusyak et al. (2024), separating pre-event trend testing from post-event treatment effect estimation. The results resemble those of the event study method, as shown in Figure 2: if the majority of pre-event estimated coefficients are insignificant, parallel trends can be deemed established. (2024). This approach separates the ex ante parallel trend test from the post-treatment effect estimation. Similar to the event study method, the results indicate that if the majority of ex ante estimated coefficients are insignificant, it can be concluded that there is no significant trend difference between the treatment and control groups prior to the official account launch. This provides further evidence supporting the validity of the parallel trend test.

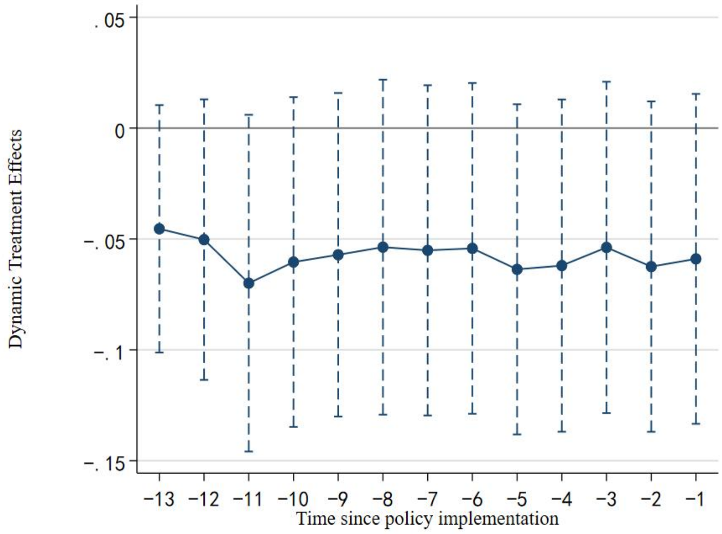


Figure 5 Robustness Test for Pre-Trends

6.4.3 Sensitivity Analysis for Parallel Trends

Event study methods focus on ex ante testing. While they possess considerable persuasive power, they have yet to directly provide testing methods and results for ex post trends within counterfactual frameworks. Drawing on Rambachan & Roth (2023)'s Honest DID approach, this paper explores the impact of policy treatment effects when parallel trend conditions may be violated. Its design principle permits some deviation in post-treatment trends (as shown in Figure 3). The Relative Magnitude (RM) assumption limits the post-treatment trend difference to no more than Mbar times the maximum pre-treatment difference (Mbar typically takes values of 0.5, 1, 1.5, or 2). Robust confidence intervals are then calculated for different Mbar values.

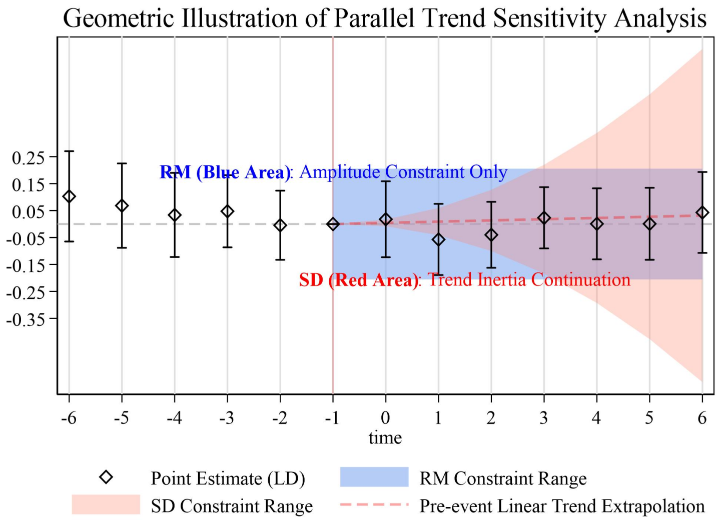


Figure 6 Sensitivity analysis

We utilized RM and SD to process sensitivity analysis and captured the maximum deviation leading to the deviation from the parallel trend. The figure 4 shows the results, indicating that both exhibit good robustness.

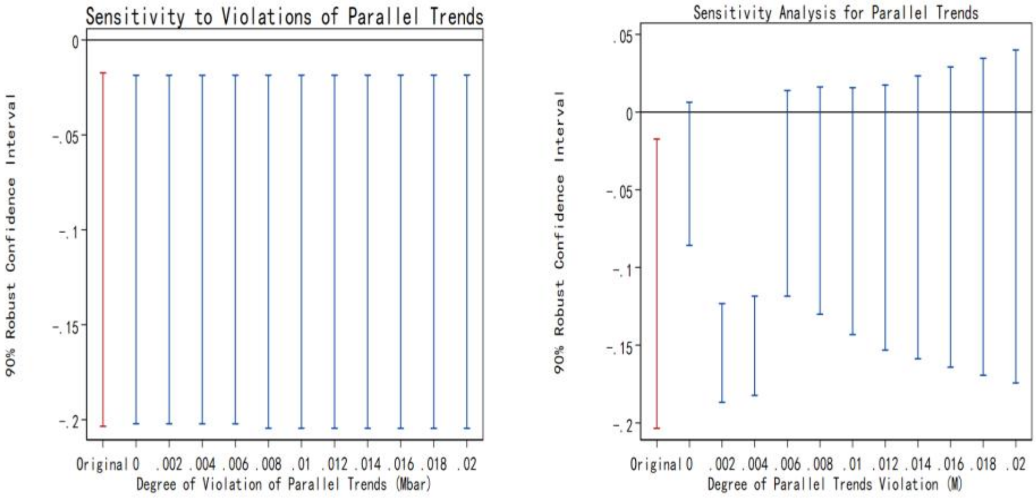


Figure 7 RM and SD tests

6.4.4 Nonlinear Violati ons and Sensitivity Analysis

In real-world policy pilots, policymakers often adopt either “cherry-picking pilots” (selecting the best-performing units) or “support-the-weak pilots” (targeting underperforming units). As a result, the treatment and control groups often exhibit group-specific time trends, leading to divergent trends in the outcome variable before and after the policy shock. This divergence violates the parallel trends assumption. According to Roth (2022), conventional event-study models based on two-way fixed effects are susceptible to truncation error, which prevents the model from accurately identifying potential violations of the parallel trend assumption between the treatment and control groups. Therefore, we conduct a test for nonlinear violations of the parallel trends assumption on the dependent variables to examine the efficacy of the event-study approach in the presence of potential nonlinear violations. We conduct the aforementioned test by customizing the Degree of Nonlinear Violation (DNV) in the pre-trends. To comprehensively account for various scenarios of nonlinear violations, the results are shown in Table 2.

Table 3 Possible linear and nonlinear adverse trends

Statistic	Mismatch	
r(Power)	0.9980	0.9952
r(Bayes)	0.4154	0.3925
r(LR)	0.0144	0.8636
Linear	0.05	0.8636
Non-linear		0.009

Given linear and nonlinear deviation levels of 0.05 and 0.009 respectively, both r(LR) and r(Bayes) approach 0, while r(power) exceeds 0.9.[5] This indicates the high effectiveness

5 r(LR) likelihood ratio test measures the similarity between observed data and the hypothesized adverse trend; ideally, it should be as close to 0 as possible. r(Bayes) represents the Bayesian test, indicating the probability ratio of the true adverse trend relative to the ideal parallel trend when the pre-trend test passes. A smaller value is better. Ideally, it should be less than 0.1. r(Power) is the most critical metric for researchers to focus on. This indicator represents the “sensitivity” of the event study method, meaning the probability that the event study model designed by the researcher can capture the adverse trend if it truly exists.

of the previously established event method, supporting the reliability of this study's design.

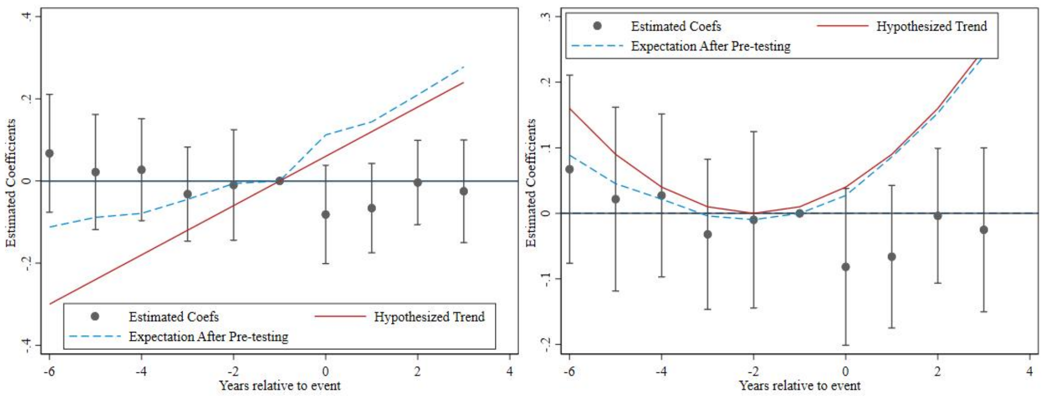


Figure 8 Test for Nonlinear Violations of the Parallel Trends Assumption

Figure 5 serves as a supplementary visual for the efficacy analysis, intuitively illustrating the fitted results of the hypothetical negative trend against the event study model. The solid red line represents the hypothesized negative trend, while the dashed blue line indicates the estimated mean outcome across periods assuming the negative trend exists and prior tests are non-significant. When assuming a linear bad trend, the blue dashed line partially overlaps within the light red confidence interval. When assuming a quadratic trend, the overlap between the blue dashed line and the light red confidence interval is lower, indicating a relatively higher likelihood of a linear bad trend. However, given the high r (Power) indicator of 0.99 and the small magnitude of parameter violations (all < 0.05), the parallel trend tests conducted earlier in this paper remain reasonably reliable.

6.4.5 Robust Estimator for Heterogeneous Treatment Effects

The estimation of the multi-period difference-in-differences (DID) model is essentially a weighted average of the treatment effects across different cohorts and time periods, which implicitly assumes homogeneous treatment effects. However, in reality, different individuals may respond heterogeneously to policy shocks. Therefore, the varying intensity of firms' reactions to the staggered rollout of tax-related public accounts could potentially introduce bias into the estimates. To address this concern, this paper employs multiple robust estimators for validation.

Specifically, we utilize the did_multiplegt_dyn command to test for the presence of heterogeneous treatment effects. The results indicate that treatments with negative weights accounted for 1.05% of all treatment effects, with a total negative weight of -0.0105. This demonstrates significant heterogeneity in treatment effects, rendering the TWFE estimator's results unstable. Therefore, the cluster-time weighted average method is employed for estimation.

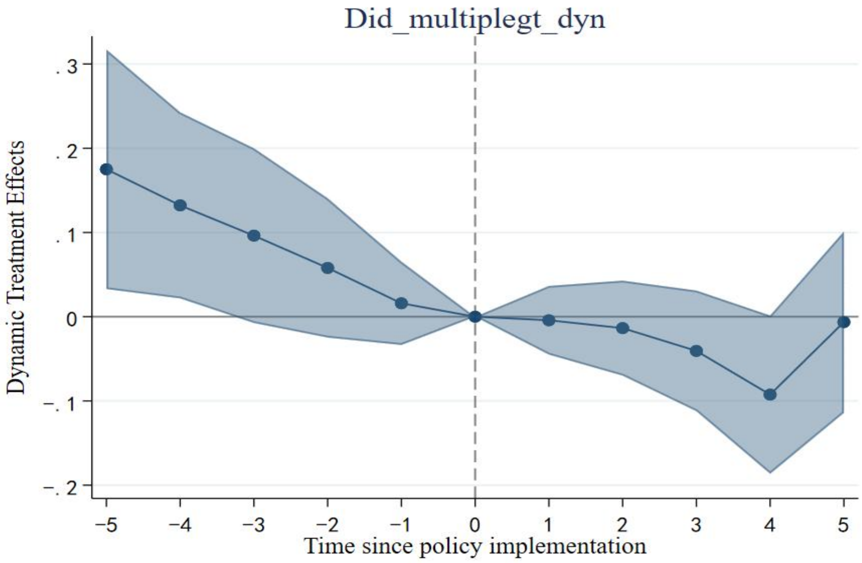


Figure 9 Test for Heterogeneous Treatment Effects

6.6 Robustness checks and endogeneity test[6]

6.6.1 Alternative fixed effects and models for expected data assets

To enhance the robustness of results, we follow Ghaly et.al.(2020). We employed the word2vec deep learning algorithm to calculate keywords related to data assets in annual reports and incorporated it into Eq (17) to predict Exp_data precisely. We also incorporated additional industrial fixed effects to forecast Exp_data and then calculated Mismatc (mismatch) . The results are shown in Table 3, the coefficient of IDC is still significantly negative in column (1) and column (2). And in scenarios of data hoarding, the coefficient for IDC is significantly negative, whereas it is not significant in data poverty. This aligns with the results from the benchmark regression, indicating robustness of the findings.

Table 4 Alternative fixed effects and models for expected data assets

Variable	(1) mismatch	(2) mismatch	(3) mismatch	(4) mismatch
IDC	-0.0794*	-0.0838*	-0.116***	-0.0737
	(0.0400)	(0.0496)	(0.0363)	(0.0760)
_cons	0.00239	-0.663**	0.784***	-0.00276
	(0.00147)	(0.319)	(0.298)	(0.311)
Controls	No	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	38639	38639	18816	18592
adj. R2	0.503	0.504	0.533	0.142

[6] Other robustness and endogeneity tests provided in the appendix

6.1.2 Excluding other policy interference

We followed Xu nuo et.al (2025), considering that some listed companies primarily operate as cloud service providers, holding an IDC operating license is not primarily aimed at empowering the real economy through computing power. Therefore, it is necessary to avoid estimation biases generated by cloud service providers. On one hand, major domestic cloud service providers (such as Alibaba Cloud Computing Co., Ltd., Tencent Cloud Technology Co., Ltd., and Huawei Cloud Computing Technology Co., Ltd.) all belong to the software and information technology services sector, samples classified under Industry I65 in the " Guidelines for Industry Classification of Listed Companies" (2012 Revision) were excluded. On the other hand, 18 cloud service providers were comprehensively identified by analyzing the listed companies' business scopes, primary products, and management discussions and analyses, and samples were further excluded based on this identification. We also excluded other data policies, like data trading and data governing[7].

Table 4 shows the results. Column (1) includes data trading policy; column (2) includes data governing policy; column (3) excludes the firms in Software and Information Technology Services; column (4) excludes cloud service provide companies and column (5) excludes all these factors that may influence regression results. The coefficient of IDC is still significantly negative and the coefficients of trade and govern are not significant, indicating the results are robust.

Table 5 Excluding other policy interference

Variable	(1) Mismatch	(2) Mismatch	(3) Mismatch	(4) Mismatch	(5) Mismatch
trade	-0.0156				-0.0211
	(0.0169)				(0.0184)
Govern		0.00149			0.0164
		(0.0198)			(0.0214)
IDC			-0.182***	-0.103**	-0.178***
			(0.0604)	(0.0515)	(0.0613)
_cons	-0.526*	-0.528*	-0.345	-0.495	-0.327
	(0.316)	(0.316)	(0.323)	(0.314)	(0.323)
Controls	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
N	38639	38639	36637	38536	36589
adj. R2	0.514	0.514	0.447	0.511	0.446

Note: Columns (3) excludes the firms in Software and Information Technology Services and column (4）excludes cloud service provide companies. So, the number of observations declines.

6.1.3 Instrumental variable (IV) analysis

A key challenge in our analysis is the potential endogeneity of IDC, which may arise from several sources. First, omitted variable bias could exist if unobserved factors, such as

[7] These policies ask companies to control data quality precisely which may affect the outcome.

regional digital conditions and management decisions, influence both the likelihood of gaining an IDC license and data asset mismatch. Second, reverse causality is a concern, Enterprises with lower levels of data asset mismatch (i.e., more rational data investment decisions) have greater motivation to apply for IDC licenses to further consolidate their competitive edge or meet compliance requirements. Alternatively, regulatory authorities may favor granting licenses to enterprises whose data investment practices are more prudent and better aligned with industrial policy objectives during the licensing approval process. Additionally, selection bias may exist. The issuance of IDC licenses is not randomly assigned. Typically, larger enterprises with robust digital foundations, strong management capabilities, or more standardized data management practices are more likely to proactively apply for and successfully obtain licenses. These companies already possess higher levels of data governance, and their degree of data asset mismatch—whether hoarding or scarcity—may inherently be lower.

To further address endogneity, we implement an IV approach. Due to the peer effect, the industry in which a firm operates exerts a significant influence on its own decision-making. This paper employs the proportion of AI computing centers among other firms in the same industry during the same year as the first instrumental variable (IV1). Building upon existing research (Manacorda and Tesei, 2020), we employ the lightning strike frequency of a firm's city of operation as an instrumental variable for its cloud adoption. Lightning strike frequency is defined as the number of lightning strikes per square kilometer annually in that city. Since lightning strike frequency is a constant that does not vary over time, directly incorporating it into the model regression would result in collinearity with the firm-specific effects. Therefore, in the estimation process, the lightning strike frequency is multiplied by the growth rate of national long-distance optical cable lines to obtain the instrumental variable (IV2).

In terms of causality, lightning strikes directly impact data center physical security and operational costs. High lightning frequency significantly increases the risk of power surges and equipment failure rates, compelling enterprises to avoid such regions during site selection and substantially raising investments in lightning protection infrastructure (Greenberg et al., 2022). Consequently, cities with higher lightning density exhibit lower probabilities of enterprises establishing smart computing centers, satisfying the correlation requirements for instrumental variables. Regarding exogeneity, lightning frequency is determined by purely natural conditions such as atmospheric circulation and geographical topography, with no reverse causality to local economic development, industrial policies, or firm-level micro-decisions (Acemoglu et al., 2001). To further ensure the exclusion constraint.

Table 5 shows the results of instrument variable regression. Column (1) shows that the regression coefficient for IV1 passed the significance test at the 1% level, and the weak instrumental variable test was passed. Column (2) indicates that the regression of IDC is significant negative. Column (3) and (4) also shows the IV2 can get the similar results.

Table 6 Instrument variable regression

Variable	(1) First IDC	(2) Second Mismatch	(1) First IDC	(2) Second Mismatch
IDC		-2.667*		-1.626*
		(1.404)		(0.903)
IV1	-0.524**			
	(0.213)			
IV2			-0.473*	
			(0.213)	
Kleibergen-Paap rk Wald F statistic	26.430		16.718	
10% maximal IV size	16.38		16.38	
Controls	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Obs	32436	32436	32436	32436

7. Channel analysis

7.1 Cost reduction and efficiency improvement

To further elucidate the intrinsic mechanism by which IDCs mitigate data asset mismatches, this paper examines two dimensions: data factor utilization efficiency and data asset structural optimization. Theoretically, as the physical carrier and computing hub for corporate data resources, IDC construction and operation not only enhance the economies of scale in data storage and processing but also significantly improve the utilization level of data elements—that is, the economic value generated per unit of data assets—through intensive management and technological empowerment. This efficiency gain directly manifests as elevated data utilization levels, forming a “cost-reduction” pathway to alleviate mismatches. Simultaneously, IDC adoption drives a dual evolution of corporate data assets: on one hand, enterprises leverage stable computing infrastructure to accelerate transforming raw data into proprietary data assets that support internal decision-making and optimize business processes, thereby strengthening data-driven operational capabilities; on the other hand, by enhancing data governance standards and compliance certifications, enterprises can market standardized, high-value data products as transactional data assets, enabling external empowerment and value realization of data resources. The accumulation of these two types of data assets corresponds to the dual internal and external cycles of data “quality enhancement” and “efficiency improvement.” This provides micro-level evidence that improving efficiency and activating assets are key to governing data asset misallocation.

Drawing on Ganotakis et al. (2023) and Cohn et al. (2022), we employ text analysis of corporate annual reports to construct our key variables. First, to assess data element utilization, we measure the disclosure frequency of five indicators—artificial intelligence, blockchain, cloud computing, big data technology, and big data applications. Using the sum of these frequencies, we derive a metric for data element utilization. Second, to categorize data assets, we use Word2Vec with "information," "network," "digital," and "data" as seed terms to identify related keywords. Based on specific usage contexts, we classify data assets into On-Premises Data Assets (ODA) and Transaction-Based Data Assets (DDA). ODA encompasses keywords such as digital infrastructure, digital factories, digital equipment, and the digital economy, reflecting internal operational use. DDA includes keywords such as digital platforms, digital trade, digital certification, and digital consumption, indicating data assets prepared for external transactions. For each firm-year, we calculate the frequency of these keywords, take the proportion relative to total text vocabulary, and apply a logarithmic transformation to address right skewness. This process yields continuous variables measuring the degree of corporate data assetization along both dimensions.

The results are presented in Table 7. The regression coefficients for IDC are all significantly positive, indicating that IDC significantly enhance enterprises' utilization efficiency of data elements while simultaneously increasing the accumulation of both self-use and transactional data assets. This finding provides direct empirical support for the cost-reduction and efficiency-enhancement mechanism. As an intensive computing infrastructure, IDC reduces the unit costs of enterprise data storage, processing, and maintenance through economies of scale and technological empowerment. This enables higher economic value from the same data investment, directly reflected in improved data element utilization. Furthermore, IDC adoption drives deeper development of corporate data assets. The growth of self-use data assets reflects enterprises internalizing data resources as core production factors supporting operations and decision-making, achieving efficiency improvements through data-driven processes. The accumulation of transactional data assets indicates that enterprises leverage IDCs' compliance frameworks and technical capabilities to productize high-value data, opening external channels for data monetization. Collectively, these three effects constitute the "cost reduction and efficiency enhancement" pathway through which IDC mitigate data asset misallocation. Improved data element utilization directly optimizes existing resource allocation and curbs inefficient hoarding; growth in self-use assets guides data back to its business origins, reducing idle waste; and expansion of transactional assets releases data value through market mechanisms, indirectly lowering enterprises' hoarding incentives. These results confirm that IDCs primarily curb hoarding by guiding enterprises to activate existing data resources and enhance utilization efficiency, rather than by simply stimulating incremental investment.

Table 7 Cost reduction and efficiency improvement

Variable	(1) Utilization	(2) ODA	(3) DDA
IDC	0.389***	0.382***	0.417***
	(0.0461)	(0.0444)	(0.0629)
_cons	0.985***	0.997***	-1.663***
	(0.234)	(0.227)	(0.335)

Controls	Yes	Yes	Yes
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
N	38639	38639	38639
adj. R2	0.737	0.725	0.693

7.2 Resource Optimization

The scalable computing power and stable data environment provided by IDC lay a solid foundation for enterprises to pursue data-driven technological innovation. Specifically, the introduction of IDC will drive resource optimization across three dimensions: First, the enhancement of computing infrastructure lowers the technical barriers for enterprises to perform data processing, model training, and algorithm iteration, directly boosting the output of digital technology patents. As a direct output of technological innovation, this reflects the process by which enterprises transform data resources into core technological capabilities. Second, leveraging data hubs and intelligent analytics platforms built on IDC infrastructure enables enterprises to extract deeper data value and enhance digital integration depth—the degree to which data elements are embedded in core business processes, reflecting the quality and complexity of technology application. Third, expanded computing power allows enterprises to apply data technologies more broadly across diverse scenarios such as production, operations, and marketing, thereby broadening digital integration scope—the coverage and penetration rate of technology application. Patent output establishes the technological foundation, application depth enhances implementation quality, and application breadth unleashes economies of scale. Collectively, these factors optimize the allocation efficiency of corporate data assets and mitigate data asset mismatches. IDC primarily achieves this by empowering technological innovation and deep application within enterprises, guiding hoarders to transform existing data resources into tangible technical capabilities rather than merely increasing investment scale.

We match the primary classification codes of listed companies' patents obtained from the National Intellectual Property Administration with the official “Statistical Classification of the Digital Economy and Its Core Industries 2021” released by the National Bureau of Statistics. This process will yield the number and proportion of digital patent applications filed by listed companies each year. Adding 1 and taking the natural logarithm will generate an indicator measuring corporate digital technology innovation (patent). The breadth of digital technology application refers to the technical support required for enterprises to collect, store, and process massive amounts of data, such as big data, cloud computing, blockchain, and so on. The depth of digital technology application encompasses digital management and digital production. Digital management refers to the application of digital technologies to achieve intelligent management practices in organization, production, sales, and services. Digital production denotes the enhancement of output and efficiency in traditional manufacturing through digital technologies, emphasizing the integration of digital technologies with the real economy. We employed text analysis methodology by searching, matching, and conducting word frequency statistics on keywords related to “digital technology” within listed companies' annual reports, categorized according to the breadth and depth of digital technology application.

Table 8 shows the results. The coefficients for IDC on digital patents, digital citation depth, and digital citation breadth are all significantly positive, indicating that IDC significantly promotes enterprises' technological innovation output while simultaneously enhancing the depth of digital technology integration into core business processes and the breadth of application scenario coverage. This finding provides direct empirical support for resource optimization mechanisms. The introduction of IDC drives resource optimization through three dimensions: First, stable and efficient computing power supply lowers the technical barriers for enterprises to conduct data processing, model training, and algorithm iteration, directly boosting the output of digital technology patents. This reflects the "technology sedimentation" process where enterprises transform data resources into core technological capabilities. Second, leveraging data hubs and intelligent analytic platforms built on IDC enables enterprises to extract deeper data value, enhancing digital adoption depth—the integration of data elements into core production, operational, and decision-making processes, reflecting the quality and complexity of technology application. Third, expanded computing power enables enterprises to apply digital technologies more broadly across diverse scenarios such as production, marketing, and supply chains, thereby broadening the scope of digital adoption—measured by the coverage and cross-departmental penetration of technology applications. These three effects constitute the "resource optimization" pathway through which IDC mitigate data asset misallocation: patent output growth signifies the accumulation of enterprise data technology capabilities, providing core momentum for resource optimization; enhanced application depth signifies the upgrade of data elements from "supporting tools" to "core production factors," driving intensive utilization of existing resources; and expanded application breadth unleashes the latent value of data assets through the scale effects of technology deployment. Consistent with the earlier group regression results, this resource optimization mechanism is more pronounced in the overinvestment group. This indicates that IDC primarily empowers enterprises to innovate and deepen technological applications, guiding hoarders to transform existing data resources into substantive technological capabilities rather than merely increasing investment scale. This finding reveals an "empowerment-based" approach to addressing data asset misallocation—technological innovation, not regulatory constraints, is the long-term mechanism for achieving optimal resource allocation.

Table 8 Resource Optimization

Variable	(1) patent	(2) depth	(3) width
IDC	0.0346*	0.600***	0.831***
	(0.0185)	(0.0833)	(0.0853)
_cons	-0.203	-4.027***	-6.961***
	(0.126)	(0.406)	(0.482)
Controls	Yes	Yes	Yes
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes
N	38639	38639	38639
adj. R2	0.721	0.753	0.752

8. Heterogeneity Analysis

8.1 Digital talents

An organization's internal digital governance capabilities are the critical prerequisite for IDC to fulfill its regulatory and enabling functions. Enterprises with executives possessing digital backgrounds or those that have established roles such as Chief Information Officer (CIO), Chief Data Officer (CDO), or Chief Technology Officer (CTO) demonstrate a deeper understanding of data strategy. They better grasp the strategic value of IDC in optimizing data governance and intensifying resource utilization, thereby more effectively transforming IDC's compliance constraints into an optimization driver for data investment decisions.

This study adopts the methodology of Benlian et al. (2018), constructing a dummy variable for executives' digital background using their academic major information from personal characteristic data of listed company directors, supervisors, and senior management. Executives whose majors involve “information, intelligence, software, electronics, communications, systems, networks, automation, wireless, or computer” are classified as having a digital background(Background). This indicator is coded as 1 if any executive in the listed company possesses a digital background, and 0 otherwise. Additionally, using the CSMAR database, we examine whether the company has established positions such as “Chief Digital Officer,” “Chief Information Officer,” or “Chief Technology Officer.” If any of these digital roles exists(Talent), the indicator is assigned a value of 1; otherwise, it is 0. Separate regression analyses are conducted for each group.

The results are shown in table 9. The mitigating effect of IDC on data asset mismatch is only significant in the group of enterprises lacking digital executive backgrounds and without a Chief Information Officer or Chief Data Officer. It is not significant in the group possessing these digital governance characteristics. This finding reveals the “shortfall-filling” effect of IDC as an exogenous regulatory tool. For enterprises already possessing digital governance capabilities, their data investment decisions are relatively standardized, and their data governance systems are well-established. The marginal governance improvements brought by IDC implementation are limited, rendering them statistically insignificant. Conversely, enterprises with weak digital governance often lack effective data investment decision-making mechanisms, making them more prone to blind hoarding or herd investment behaviors. IDC's compliance constraints and infrastructure empowerment precisely fill the governance gaps in such enterprises.

Table 9 Digital Talents

Variable	(1) Mismatch Background=0	(2) Mismatch Background=1	(3) Mismatch Talent=0	(4) Mismatch Talent=1
IDC	-0.144**	0.0416	-0.123**	0.400
	(0.0610)	(0.100)	(0.0524)	(0.290)
_cons	-0.497	-1.080	-0.544*	12.32

	(0.333)	(1.503)	(0.321)	(9.184)
Controls	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	35080	2679	37694	216
adj. R2	0.489	0.674	0.518	0.562

8.2 Technological endowment

A company's technological foundation determines its capacity to adopt and utilize IDC. Digital enterprises and high-tech enterprises inherently exhibit active data activities and deep technological accumulation. The introduction of IDC can create synergies with their existing technological systems, accelerating the transformation of data elements into core production capabilities.

Following the National Bureau of Statistics' "Statistical Classification of the Digital Economy and Its Core Industries (2021)" and the "Guidance on Industry Classification for Listed Companies" by the China Association of Listed Companies, we classify listed companies operating in the following sectors as digital enterprises (Digital): computer, communications, and other electronic equipment manufacturing; telecommunications, broadcasting, and satellite transmission services; internet and related services; and software and information technology services. All others are categorized as traditional enterprises. Additionally, based on the High-Tech Enterprise Certification Network's filing announcements, enterprises classified as high-tech (Tech) enterprises are assigned a value of 1, while others receive a value of 0. Separate regression analyses are conducted for each group.

The results are shown in table 10. This study finds that the mitigating effect of IDC on data asset mismatch is significant only in non-digital enterprises and non-high-tech enterprises, but not in digital enterprises and high-tech enterprises. This result indicates that the governance effect of IDC exhibits distinct "filling gaps" characteristics. For enterprises with weak technological endowments, their data investment decisions often lack support from mature technical frameworks, making them more prone to falling into the trap of blind accumulation or herd investment. The introduction of IDC precisely fills this technological gap. On one hand, the centralized computing infrastructure lowers the technical barriers for data collection, storage, and processing, mitigating disorderly investments caused by insufficient technical capabilities. On the other hand, the compliance and energy consumption standards embedded within IDC impose external constraints, guiding these enterprises to establish basic data governance norms. In contrast, digital enterprises and high-tech companies already possess robust data technology capabilities and mature governance systems, making their data investment behavior relatively rational. The marginal improvements brought by IDC are limited for these entities.

Table 10 Technological endowment

Variable	(1) Mismatch Digital=0	(2) Mismatch Digital=1	(3) Mismatch Tech=0	(4) Mismatch Tech=1
IDC	-0.248**	0.00136	-0.274***	0.0857
	(0.0977)	(0.0577)	(0.0781)	(0.0587)
_cons	-0.194	-1.031**	-0.435	-1.266
	(0.522)	(0.419)	(0.354)	(0.775)
Controls	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	15553	22325	31764	6137
adj. R2	0.424	0.563	0.405	0.636

8.3 Location Conditions

The operational efficiency of IDCs is constrained by regional energy endowments and digital infrastructure. Energy consumption levels directly impact IDC operating costs; enterprises in high-energy-consumption regions face stronger cost constraints, making marginal gains in cost reduction and efficiency enhancement more pronounced. Meanwhile, the scale of telecommunications service investment reflects the sophistication of regional digital infrastructure. Enterprises in areas with superior infrastructure can access IDC resources more readily and enjoy lower network latency, thereby amplifying the resource allocation optimization role of IDCs.

Based on regional statistical yearbooks, we grouped regions into high and low categories using the median values of industrial electricity consumption (Energy) and telecommunications investment (Foundation) for each region, then performed grouped regression analysis. Values above the median were assigned a score of 1, while those below received a score of 0.

The results are shown in table 11. This study finds that the inhibitory effect of IDC on data asset mismatch is significant in the high-energy consumption group and the low-telecom investment group, but not in the low-energy consumption group and the high-telecom investment group. This result reveals two complementary mechanisms through which IDC exert their influence. On one hand, in high-energy consumption regions, enterprises face stronger energy cost constraints and environmental compliance pressures. As high-energy-consuming facilities, the introduction of IDC significantly increases operational costs for enterprises, thereby imposing strong cost constraints on the indiscriminate expansion of data assets and effectively curbing data hoarding. This demonstrates the governance effect of IDC through a “cost-driven” mechanism. On the other hand, in regions with low telecommunications investment, digital infrastructure is relatively underdeveloped. Enterprises are constrained by limited network bandwidth and data transmission capabilities, resulting in low levels of data application. The introduction of IDC directly addresses local infrastructure deficiencies by providing centralized computing power support. This significantly lowers the technical barriers to data operations, enabling enterprises that previously underinvested or invested in disorderly ways due to infrastructure constraints to optimize their data resource allocation. This reflects the “infrastructure empowerment” effect of IDC.

Table 11 Location Conditions

Variable	(1) Mismatch Energy = 0	(2) Mismatch Energy = 1	(3) Mismatch Foundation = 0	(4) Mismatch Foundation = 1
IDC	-0.289***	0.0104	-0.140	-0.131**
	(0.0811)	(0.0651)	(0.105)	(0.0618)
_cons	-0.602	-0.124	-0.0775	-0.471
	(0.461)	(0.463)	(0.454)	(0.543)
Controls	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	17446	16780	17163	16961
adj. R2	0.488	0.562	0.514	0.549

9. Conclusion

This paper investigates the formation and governance of data asset mismatch from both theoretical and empirical perspectives. We first develop a dynamic theoretical framework in which data capital functions as informational capital that improves firms’ knowledge precision through Bayesian learning. Within this framework, firms jointly determine investment in physical capital and data infrastructure while facing adjustment costs in data accumulation. The theoretical analysis shows that the optimal level of data accumulation depends on the balance between the marginal informational benefits of data and the user cost of maintaining data capital. Under different technological conditions, the economy may converge to heterogeneous equilibria characterized by either “data poverty” or “data hoarding”.

To test the theoretical implications, we further conduct an empirical analysis using firm-level panel data and exploit the acquisition of Internet Data Center (IDC) licenses as a quasi-natural experiment. The DID results show that IDC licenses significantly alleviate data asset mismatch within firms. This finding suggests that improvements in computing infrastructure play an important role in facilitating the efficient allocation and utilization of data resources.

Mechanism analyses reveal two primary channels through which IDC infrastructure mitigates data asset mismatch. First, IDC infrastructure reduces the cost of data storage and processing and significantly improves the utilization efficiency of data elements. At the same time, it promotes the accumulation of both self-use and transactional data assets, indicating that enterprises are able to better internalize and monetize their data resources. Second, IDC infrastructure strengthens firms' digital innovation capacity. The empirical results show that IDC licenses significantly increase digital patent output as well as the depth and breadth of digital technology adoption, suggesting that enhanced computing power enables firms to transform data resources into technological capabilities and expand the application scope of digital technologies.

Overall, the findings highlight the important role of computing infrastructure in shaping firms' data investment decisions and improving the efficiency of data asset allocation. By integrating theoretical modeling with empirical evidence, this study provides a systematic explanation for the formation of data asset mismatch and offers new insights into how digital infrastructure development can promote the productive utilization of data resources in the digital economy.

Reference

Brynjolfsson, E., Rock, D., & Syverson, C. (2021). The productivity J-curve: How intangibles complement general purpose technologies. American Economic Journal: Macroeconomics, 13(1), 333-372.

Hall, R. E. (2004). Measuring factor adjustment costs. The Quarterly Journal of Economics, 119(3), 899-927.

Bloom, N. (2009). The impact of uncertainty shocks. econometrica, 77(3), 623-685.

Jovanovic, B. (1982). Selection and the Evolution of Industry. Econometrica: Journal of the econometric society, 649-670.

Veldkamp, L. (2023). Valuing data as an asset. Review of Finance, 27(5), 1545-1562.

Jones, C. I., & Tonetti, C. (2020). Nonrivalry and the economics of data. American economic review, 110(9), 2819-2858.

Autor, D., Dorn, D., Katz, L. F., Patterson, C., & Van Reenen, J. (2020). The fall of the labor share and the rise of superstar firms. The Quarterly journal of economics, 135(2), 645-709.

Monteiro, P., & Suleymanov, M. (2025). Government contracts and labor investment efficiency. Journal of Corporate Finance, 92, 102771.

Farboodi, M., & Veldkamp, L. (2021). A model of the data economy (No. w28427). Cambridge, MA, USA: National Bureau of Economic Research.

Cong, L. W., Xie, D., & Zhang, L. (2021). Knowledge accumulation, privacy, and growth in a data economy. Management science, 67(10), 6480-6492.

Lo, A. W., & MacKinlay, A. C. (1990). Data-snooping biases in tests of financial asset pricing models. The Review of Financial Studies, 3(3), 431-467.

Chen, L., Pelger, M., & Zhu, J. (2024). Deep learning in asset pricing. Management Science, 70(2), 714-750.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. Journal of econometrics, 225(2), 254-277.

Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. Journal of econometrics, 225(2), 200-230.

Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). Synthetic difference-in-differences. American economic review, 111(12), 4088-4118.

Gormley, C. J., & Gormley, S. J. (2012). Data hoarding and information clutter: The impact on cost, life span of data, effectiveness, sharing, productivity, and knowledge management culture. Issues in Information Systems, 13(2), 90-95.

Mitchell, A. (2012). From data hoarding to data sharing. Journal of Direct, Data and Digital Marketing Practice, 13(4), 325-334.

McKellar, K., Sillence, E., Neave, N., & Briggs, P. (2024). Digital accumulation behaviours and information management in the workplace: exploring the tensions between digital data hoarding, organisational culture and policy. Behaviour & Information Technology, 43(6), 1206-1218.

Birch, K. (2023). Data assets. In *Data enclaves* (pp. 41-59). Cham: Springer Nature Switzerland.

Einav, L., & Levin, J. (2014). Economics in the age of big data. *Science*, *346*(6210), 1243089.

Jones, C. I., & Tonetti, C. (2020). Nonrivalry and the economics of data. *American economic review*, *110*(9), 2819-2858.

Griliches, Z. (1979). Sibling models and data in economics: Beginnings of a survey. *Journal of political Economy*, *87*(5, Part 2), S37-S64.

Tella, R. D., & MacCulloch, R. (2006). Some uses of happiness data in economics. *Journal of economic perspectives*, *20*(1), 25-46.

Bergemann, D., Bonatti, A., & Gan, T. (2022). The economics of social data. *The RAND Journal of Economics*, *53*(2), 263-296.

Nayak, B. S., & Walton, N. (2024). The future of platforms, big data and new forms of capital accumulation. *Information Technology & People*, *37*(2), 662-676.

Bond, S., Leblebicioğlu, A., & Schiantarelli, F. (2010). Capital accumulation and growth: a new look at the empirical evidence. *Journal of Applied Econometrics*, *25*(7), 1073-1099.

Bichler, S., & Nitzan, J. (2015). Capital accumulation: fiction and reality. *Real-World Economics Review*, (72, September), 47-68.

Chang, Q., Wu, M., & Zhang, L. (2024). Endogenous growth and human capital accumulation in a data economy. *Structural Change and Economic Dynamics*, *69*, 298-312.

Acemoglu, D., & Restrepo, P. (2018). The race between man and machine: Implications of technology for growth, factor shares, and employment. *American economic review*, *108*(6), 1488-1542.

Admati, A. R., & Pfleiderer, P. (1990). Direct and indirect sale of information. *Econometrica: Journal of the Econometric Society*, 901-928.

Aghion, P., Bergeaud, A., Boppart, T., Klenow, P. J., & Li, H. (2023). A theory of falling growth and rising rents. *Review of Economic Studies*, *90*(6), 2675-2702.

Agrawal, A., McHale, J., & Oettl, A. (2018). Finding needles in haystacks: Artificial intelligence and recombinant growth. In *The economics of artificial intelligence: An agenda* (pp. 149-174). University of Chicago Press.

Yao, L., & Qiu, Y. (2024). Market liquidity, credit maturity structure and asset mismatch in manufacturing firms. *Finance Research Letters*, *63*, 105400.

Hu, M., & Li, B. (2024). Potential risks in the easy market access: A perspective from corporate asset-debt maturity mismatch. *Research in International Business and Finance*, *70*, 102324.

Angeletos, G. M., Hellwig, C., & Pavan, A. (2006). Signaling in a global game: Coordination and policy traps. *Journal of Political economy*, *114*(3), 452-484.

Asriyan, V., Laeven, L., & Martin, A. (2022). Collateral booms and information depletion. *The Review of Economic Studies*, *89*(2), 517-555.

Atkeson, A., & Kehoe, P. J. (2005). Modeling and measuring organization capital. *Journal of political Economy*, *113*(5), 1026-1053.

Bajari, P., Chernozhukov, V., Hortaçsu, A., & Suzuki, J. (2019, May). The impact of big data on firm performance: An empirical investigation. In *AEA papers and proceedings* (Vol. 109, pp. 33-37). 2014 Broadway, Suite 305, Nashville, TN 37203: American Economic Association.

Beraja, M., Yang, D. Y., & Yuchtman, N. (2023). Data-intensive innovation and the state: Evidence from AI firms in China. *The Review of Economic Studies*, *90*(4), 1701-1723.

Eeckhout, J., & Veldkamp, L. (2022). Data and market power.

Jones, C. I., & Tonetti, C. (2020). Nonrivalry and the economics of data. *American economic review*, *110*(9), 2819-2858.

[1] Acemoglu, D., & Restrepo, P. (2018). The race between man and machine: Implications of technology for growth, factor shares, and employment. American economic review, 108(6), 1488-1542.

[2] Admati, A. R., & Pfleiderer, P. (1990). Direct and indirect sale of information. Econometrica: Journal of the Econometric Society, 901-928.

[3] Aghion, P., Bergeaud, A., Boppart, T., Klenow, P. J., & Li, H. (2023). A theory of falling growth and rising rents. Review of Economic Studies, 90(6), 2675-2702.

[4] Agrawal, A., McHale, J., & Oettl, A. (2018). Finding needles in haystacks: Artificial intelligence and recombinant growth. In The economics of artificial intelligence: An agenda (pp. 149-174). University of Chicago Press.

[5] Angeletos, G. M., Hellwig, C., & Pavan, A. (2006). Signaling in a global game: Coordination and policy traps. Journal of Political economy, 114(3), 452-484.

[6] Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). Synthetic difference-in-differences. American economic review, 111(12), 4088-4118.

[7] Asriyan, V., Laeven, L., & Martin, A. (2022). Collateral booms and information depletion. The Review of Economic Studies, 89(2), 517-555.

[8] Atkeson, A., & Kehoe, P. J. (2005). Modeling and measuring organization capital. Journal of political Economy, 113(5), 1026-1053.

[9] Autor, D., Dorn, D., Katz, L. F., Patterson, C., & Van Reenen, J. (2020). The fall of the labor share and the rise of superstar firms. The Quarterly journal of economics, 135(2), 645-709.

[10] Bajari, P., Chernozhukov, V., Hortaçsu, A., & Suzuki, J. (2019, May). The impact of big data on firm performance: An empirical investigation. In AEA papers and proceedings (Vol. 109, pp. 33-37). American Economic Association.

[11] Beraja, M., Yang, D. Y., & Yuchtman, N. (2023). Data-intensive innovation and the state: Evidence from AI firms in China. The Review of Economic Studies, 90(4), 1701-1723.

[12] Bergemann, D., Bonatti, A., & Gan, T. (2022). The economics of social data. The RAND Journal of Economics, 53(2), 263-296.

[13] Bichler, S., & Nitzan, J. (2015). Capital accumulation: fiction and reality. Real-World Economics Review, (72, September), 47-68.

[14] Birch, K. (2023). Data assets. In Data enclaves (pp. 41-59). Cham: Springer Nature Switzerland.

[15] Bloom, N. (2009). The impact of uncertainty shocks. econometrica, 77(3), 623-685.

[16] Bond, S., Leblebicioğlu, A., & Schiantarelli, F. (2010). Capital accumulation and growth: a new look at the empirical evidence. Journal of Applied Econometrics, 25(7), 1073-1099.

[17] Brynjolfsson, E., Rock, D., & Syverson, C. (2021). The productivity J-curve: How intangibles complement general purpose technologies. American Economic Journal: Macroeconomics, 13(1), 333-372.

[18] Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. Journal of econometrics, 225(2), 200-230.

[19] Chang, Q., Wu, M., & Zhang, L. (2024). Endogenous growth and human capital accumulation in a data economy. Structural Change and Economic Dynamics, 69, 298-298.

[20] Chen, L., Pelger, M., & Zhu, J. (2024). Deep learning in asset pricing. Management Science, 70(2), 714-750.

[21] Cong, L. W., Xie, D., & Zhang, L. (2021). Knowledge accumulation, privacy, and

growth in a data economy. Management science, 67(10), 6480-6492.

[22] Eeckhout, J., & Veldkamp, L. (2022). Data and market power.

[23] Einav, L., & Levin, J. (2014). Economics in the age of big data. Science, 346(6210), 1243089.

[24] Farboodi, M., & Veldkamp, L. (2021). A model of the data economy (No. w28427). Cambridge, MA, USA: National Bureau of Economic Research.

[25] Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. Journal of econometrics, 225(2), 254-277.

[26] Gormley, C. J., & Gormley, S. J. (2012). Data hoarding and information clutter: The impact on cost, life span of data, effectiveness, sharing, productivity, and knowledge management culture. Issues in Information Systems, 13(2), 90-95.

[27] Griliches, Z. (1979). Sibling models and data in economics: Beginnings of a survey. Journal of political Economy, 87(5, Part 2), S37-S64.

[28] Hall, R. E. (2004). Measuring factor adjustment costs. The Quarterly Journal of Economics, 119(3), 899-927.

[29] Hu, M., & Li, B. (2024). Potential risks in the easy market access: A perspective from corporate asset-debt maturity mismatch. Research in International Business and Finance, 70, 102324.

[30] Jones, C. I., & Tonetti, C. (2020). Nonrivalry and the economics of data. American economic review, 110(9), 2819-2858.

[31] Jovanovic, B. (1982). Selection and the Evolution of Industry. Econometrica: Journal of the econometric society, 649-670.

[32] Lo, A. W., & MacKinlay, A. C. (1990). Data-snooping biases in tests of financial asset pricing models. The Review of Financial Studies, 3(3), 431-467.

[33] McKellar, K., Sillence, E., Neave, N., & Briggs, P. (2024). Digital accumulation behaviours and information management in the workplace: exploring the tensions between digital data hoarding, organisational culture and policy. Behaviour & Information Technology, 43(6), 1206-1218.

[34] Mitchell, A. (2012). From data hoarding to data sharing. Journal of Direct, Data and Digital Marketing Practice, 13(4), 325-334.

[35] Monteiro, P., & Suleymanov, M. (2025). Government contracts and labor investment efficiency. Journal of Corporate Finance, 92, 102771.

[36] Nayak, B. S., & Walton, N. (2024). The future of platforms, big data and new forms of capital accumulation. Information Technology & People, 37(2), 662-676.

[37] Tella, R. D., & MacCulloch, R. (2006). Some uses of happiness data in economics. Journal of economic perspectives, 20(1), 25-46.

[38] Veldkamp, L. (2023). Valuing data as an asset. Review of Finance, 27(5), 1545-1562.

[39] Yao, L., & Qiu, Y. (2024). Market liquidity, credit maturity structure and asset mismatch in manufacturing firms. Finance Research Letters, 63, 105400.

[40] Nimark, K. P., & Pitschner, S. (2019). News media and delegated information choice. Journal of Economic Theory, 181, 160-196.

Appendix

Appendix 1：Detailed Derivation of the optimal Data Assets for Enterprises

Assuming consistency with Part III, this section begins by solving the Bellman equation.

$$(\Omega_t, D_t) = \max_{K_t, I_{D,t}} \left\{ \Pi(K_t, \Omega_t) - \frac{1}{2}\kappa_D I_{D,t}^2 + \frac{1}{1+r}\mathbb{E}_t[V(\Omega_{t+1}, D_{t+1})] \right\}$$

Among these, the evolution equation for state variables is:

Data accumulation：

$$D_{t+1} = (1-\delta_D)D_t + I_{D,t}$$

Knowledge update：

$$\Omega_{t+1} = \left[\rho^2\Omega_t^{-1} + \sigma_0^2\right]^{-1} + z_i h(D_t)\sigma_\epsilon^{-2}$$

Since the physical capital K_t does not affect future state variables， it constitutes a static decision. Take the first derivative with respect to K_t:

$$\frac{\partial \Pi}{\partial K_t} = \alpha P \mu(\Omega_t) K_t^{\alpha-1} - r = 0$$

$$K_t^* = \left(\frac{\alpha P_\mu(\Omega_t)}{r}\right)^{\frac{1}{1-\alpha}}$$

Take the first derivative with respect to $I_{D,t}$:

$$-\kappa_D I_{D,t} + \frac{1}{1+r}E_t\left[\frac{\partial V}{\partial D_{t+1}}\frac{\partial D_{t+1}}{\partial I_{D,t}}\right] = 0$$

$$\kappa_D I_{D,t} = \frac{1}{1+r}E_t\left[V_D^{'}(t+1)\right]$$

The marginal cost of current-period investment must equal the discounted value of the shadow price of data assets in the next period.

To determine $V_D^{'}(t+1)$， a total differential must be taken with respect to the state variable D_t:

$$V_D^{'}(t) = \frac{\partial V}{\partial D_t} = \frac{1}{1+r}E_t\left[\frac{\partial V}{\partial \Omega_{t+1}}\frac{\partial \Omega_{t+1}}{\partial D_t} + \frac{\partial V}{\partial D_{t+1}}\frac{\partial D_{t+1}}{\partial D_t}\right]$$

$$\frac{\partial \Omega_{t+1}}{\partial D_t} = z_i h^{'}(D_t)\sigma_\epsilon^{-2},$$

$$\frac{\partial D_{t+1}}{\partial D_t} = 1 - \delta_D$$

$$V_D^{'}(t) = \frac{1}{1+r}E_t\left[V_\Omega^{'}(t+1)\cdot z_i h^{'}(D_t)\sigma_\epsilon^{-2} + \kappa_D I_{D,t+1}(1-\delta_D)\right]$$

$$\underbrace{\kappa_D I_{D,t}}_{\text{Marginal costs}} = \frac{1}{1+r}E_t[\underbrace{V_\Omega^{'}(t+1)\frac{\partial \Omega_{t+1}}{\partial D_t}}_{The\ value\ of\ info} + \underbrace{\kappa_D I_{D,t+1}(1-\delta_D)}_{\text{The value of data assets}}] \qquad (1)$$

This equation indicates that corporate investment data serves to obtain two types of value: on the one hand, the informational benefit of enhancing predictive accuracy;

on the other hand, the residual value of assets that remain available for future use.

Applying the Envelope Theorem to knowledge precision Ω_t, it simultaneously appears in both the current profit function and the evolution term for the next period:

$$V_{\Omega}^{'}(\Omega_t) = \frac{\partial \Pi}{\partial \Omega_t} + \frac{1}{1+r} E_t \left[V_{\Omega}^{'}(t+1) \frac{\partial \Omega_{t+1}}{\partial \Omega_t} \right]$$

At steady state, all variables remain constant. According to $D_{t+1} = (1-\delta_D)D_t + I_{D,t}$, the investment level at steady state must exactly offset the discount:

$$I_D^* = \delta_D D^*$$

Marginal Investment Cost: $P_D = K_D I_D^*$

Knowledge precision $V_{\Omega}^{'}(\Omega_t) = V_{\Omega}^{'}(\Omega_{t+1}) = V_{\Omega}^{'}$, defining the persistence of knowledge is $M_{\Omega} = \frac{\partial \Omega_{t+1}}{\partial \Omega_t}$

Substitute the steady-state variables into the recursive equation and manipulate it：

$$V_{\Omega}^{'} = \frac{\partial Ш}{\partial \Omega} + \frac{1}{1+r} V_{\Omega}^{'} \cdot M_{\Omega}$$

$$V_{\Omega}^{'} \left(1 - \frac{1}{1+r} M_{\Omega}\right) = \frac{\partial \Pi}{\partial \Omega}$$

$$V_{\Omega}^{'} = \frac{\partial \Pi / \partial \Omega}{1 - \frac{1}{1+r} \frac{\partial \Omega_{t+1}}{\partial \Omega_t}}$$

$1 - \frac{1}{1+r} \frac{\partial \Omega_{t+1}}{\partial \Omega_t}$ This reflects the enduring nature of knowledge. As long as knowledge can be sustained into the future, its value will be amplified.

Take $V_{\Omega}^{'}$ into equation (1):

$$P_D = \frac{1}{1+r} \left[V_{\Omega}^{'} \cdot z_i h^{'}(D^*) \sigma_{\epsilon}^{-2} + P_D(1-\delta_D) \right]$$

$$P_D(1+r) = V_{\Omega}^{'} \cdot z_i h^{'}(D^*) \sigma_{\epsilon}^{-2} + P_D(1-\delta_D)$$

$$P_D(1+r-1+\delta_D) = V_{\Omega}^{'} \cdot z_i h^{'}(D^*) \sigma_{\epsilon}^{-2}$$

$$P_D(r+\delta_D) = \underbrace{z_i h^{'}(D^*) \sigma_{\epsilon}^{-2}}_{\text{Quantity of info}} \cdot \underbrace{\left(\frac{P(K^*)^{\alpha} \mu^{'}(\Omega^*)}{1 - \frac{1}{1+r} M_{\Omega}} \right)}_{\text{Marginal Value of info}}$$

Therefore, the optimal data stock D* depends on the user cost of holding data equaling the marginal benefit of the data.

Appendix2: The detailed definitions of variables

Appendix3: Other robustness and endogeneity tests

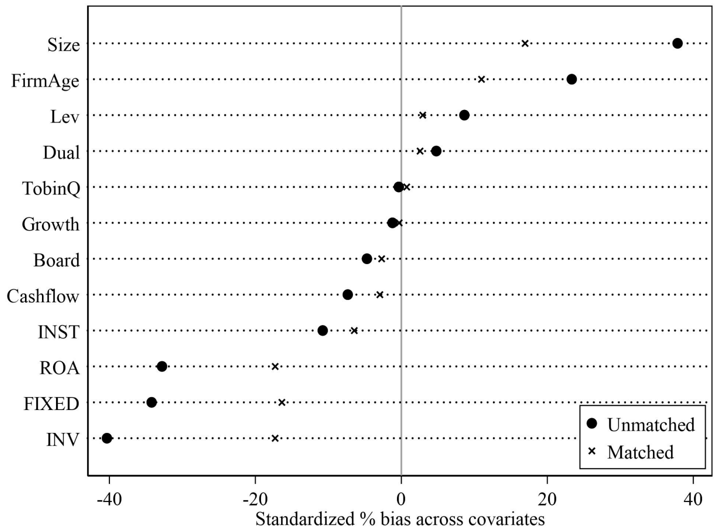
1) High-level fixed-effects regression:

To control for unobservable latent factors at the city and industry levels, this study incorporates city and industry fixed effects in addition to firm and year fixed effects. Furthermore, to account for industry-specific variations across years, the analysis also controls for the interaction between year and industry fixed effects.

The results are shown in Appendix 1, column (1) adds industrial fixed-effect and city fixed-effect; column (2) adds the year*ind fixed effect based on (1) and column (3) adds the year*city fixed effect based on (1). All the coefficient of IDC is significantly negative, indicating the results are robust.

2) PSM-DID

To mitigate the self-selection issue arising from regional resource endowment disparities affecting enterprise-certified IDCs, this study employs a kernel matching approach for PSM-DID. The matching results are presented in Appendix Figure 1, while the regression outcomes are detailed in Appendix Table 1. Both sets of results demonstrate robust estimation.



Appendix 1 Figure The results of PSM

3) Double clustering robust standard error

To control for the effects of autocorrelation in time series, this study performs dual-level cluster adjustments for standard errors at both the company and year levels.

The results are shown in Appendix 1, column (5). We clustered standard errors at both the company and year levels, the coefficient of IDC is significantly negative, indicating the regression results are robust.

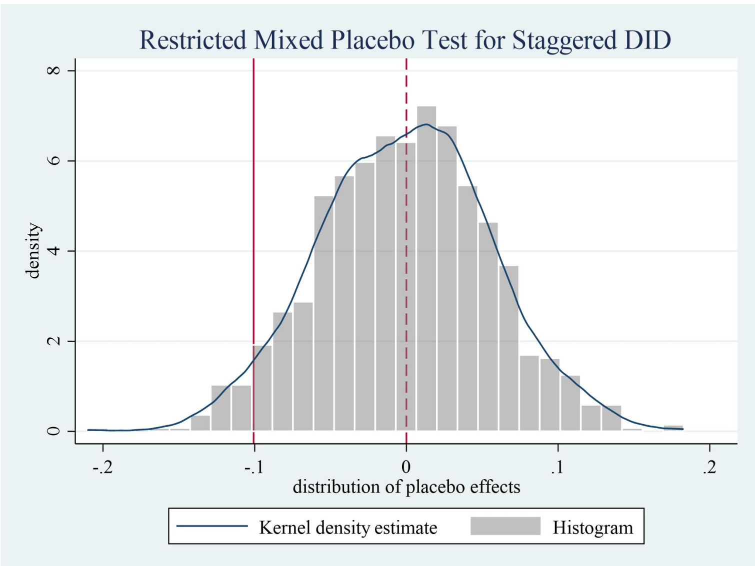
Appendix Table 2 Other robustness and endogenetiry tests

	(1)	(2)	(3)	(4)	(5)
	Mismatch	Mismatch	Mismatch	Mismatch	mismatch

IDC	-0.101**	-0.102**	-0.107**	-0.103**	-0.115**
	(0.0496)	(0.0494)	(0.0487)	(0.0507)	(0.0537)
_cons	-0.484	-0.150	-0.750**	-0.495	-0.525
	(0.318)	(0.349)	(0.325)	(0.323)	(0.341)
Controls	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Ind	Yes	Yes	Yes	No	No
City	Yes	Yes	Yes	No	No
Year*Ind	No	Yes	No	No	No
Year*City	No	No	Yes	No	No
Cluster	Stkcd	Stkcd	Stkcd	Stkcd	Stkcd year
N	37931	35928	37856	37518	37931
adj. R2	0.515	0.523	0.531	0.519	0.519

4) Placebo test:

To eliminate potential interference from unobserved random factors on the conclusions, this study repeated the regression analysis 500 times by randomly assigning the variable IDC values of 0 and 1. The distribution of the virtual regression coefficients and the differences between these virtual coefficients and the actual regression coefficients were observed. Appendix figure 2 displays the results of the placebo test, where the simulated regression coefficients exhibit a normal distribution. Although there is some overlap between the simulated and true regression coefficients, the intersection is minimal. The findings indicate that the research results are robust.



Appendix Figure 3 Placebo test